

Gender norms in agri-food systems; Adaptive Capacity



INITIATIVE ON
Gender Equality

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Context Annotated Bibliography

This annotated bibliography (AB) provides a research overview on the basis of 58 articles. The annotated bibliography was produced to support an evidence synthesis on gender norms in agri-food systems (Rietveld et al., 2023¹). The aim of the evidence synthesis was to describe how gender norms facilitate, or hamper, women's abilities to achieve economic resilience in the face of climate change challenges. For each article the title and citation are listed and a description is provided. The description covers the research topic, the methodology, the research findings and conclusions of the article. Sometimes the reviewer has added additional comments.

A total of five ABs on 'gender norms in agri-food systems' were produced, each focusing on another theme. Next to this AB on Adaptive behaviour (#4), ABs on 'Value Chains (#1)'; 'Consumer Behaviour' (#2); Food Environment (#3); and Economic Resilience (#5) were produced.

This evidence synthesis was conducted as part of Work Package One (WP 1) of the CGIAR initiative on Gender Equality (HER+). WP 1 works with a range of partners to address normative constraints that limit the capacities of women agri-food systems actors to build economic resilience to climate change challenges through developing and using gender transformative approaches.

Article selection

The Web of Science (WoS; <https://clarivate.com>) was used to systematically identify appropriate literature. The search terms in Table 1 were cross-referenced with "gender" and "women".

Local/regional/national adaptive capacity; assets; flexibility; learning; social organization; agency.
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Table 1. Search terms for 'adaptive capacity'

The following restrictions were applied: i) geographical focus on LMIC, (ii) agri-food system focus, (iii) analysis of primary data, (iv) written in the English language, (v) articles from 2016 to 2022 only. Deep reading of the thus selected articles resulted in further articles being discarded. Articles were discarded if they showed no or insufficient focus on agrifood systems, or if gender issues were not meaningfully addressed.

Author statement

Methodology and article selection (AR), article review and annotated bibliography (RV; AM), editing (RM and AR); project supervision (AR).

¹ Rietveld, A.M., Farnworth, C.R., Shijagurumayum, M., Meentzen, A., Voss, R., Morahan, R. and López, D.E. (2023). An evidence synthesis of gender norms in agrifood systems: Pathways towards improved women's economic resilience to climate change. Rome, Italy: Bioversity International. <https://hdl.handle.net/10568/136053>

Annotated bibliography

1. H Cohen, P.J., Lawless, S., Dyer, M. et al. Understanding adaptive capacity and capacity to innovate in social-ecological systems: applying a gender lens. *Ambio* 45, 309-321 (2016), <https://doi.org/10.1007/s13280-016-0831-4>

Research Topic

The objective of the paper is to understand social and gender differentiation and capacities to adapt and innovate. It analyses how gender shapes capacities in cases in the Solomon Islands, based on a qualitative study in three communities in the context of Gennovate. The article makes explicit references to social and gender norms.

Two research questions were designed as follows:

1. What socio-institutional factors shape capacity to adapt and innovate and in what ways?
2. In what ways do *social and gender norms* shape these factors and thus influence differential capacity to adapt and innovate for women and men?

Change, instability and uncertainty are inherent in social-ecological systems and influence people's ability to benefit from natural resources and improve wellbeing. Strategies to build infrastructure or provide technical innovations developed externally and delivered locally have no lasting impacts.

"Adaptive capacity is a component of both resilience and vulnerability (Adger 2006). Adaptations are the actions of individuals, communities and governments undertaken for the purpose of improving or protecting wellbeing (Adger et al. 2005). Adaptive capacity refers to the conditions that enable people to anticipate and respond to change and recover from and minimize consequences of change (Adger and Vincent 2005).

An innovation is an initiative, process or program that changes existing routines, resource flows, and may be something entirely new or a novel combination of established and new ideas. (Moore and Westley 2011)."

This article highlights how *social norms* have limiting effects not only on women but also on men, only to different degrees.

Methodology

Few studies have examined adaptive capacity and capacity to innovate and gendered differences. A framework is proposed with five dimensions of capacity to adapt and to innovate (assets, flexibility, learning, social organization, agency), which are mutually dependent.

Limits to education, physical mobility and agency meant that women and youth, in particular, felt it was difficult to establish relations with external agencies to access technical support or new information important for innovating or adapting. Willingness to bear risks and to challenge social norms hindered both women's and men's capacity to innovate, albeit to differing degrees.

The data collection included focus groups with men, women and youth and semi-structured key informant interviews. Analysis was based on a coding structure developed by Gennovate.

Relevant findings

Challenges were associated with assets (land access and quality, health and transportation services). Lack of tools or money were perceived to inhibit capacity to innovate.

Access to social, natural, physical and economic assets were socially and gender differentiated. The clearest distinctions were due to *social norms*, particularly for education, land and marine areas and cash. Higher levels of education correlated with greater access to other opportunities.

Social norms and shifts in *social norms* influenced flexibility. But gendered division of labor had become less rigid than in the past. Women had increased their workload, but more men participated in household labor than before.

Learning related strongly to capacity to innovate. Women's savings clubs, resource management committees, were not a result of environmental shocks but were interpreted as capacities to innovate rather than as adaptive capacity. Access to information, *assessment of risks and social norms* were cited as being influential on learning.

Practical modes of information delivery or training were considered to be more effective. People of lower wellbeing tended to source information and support primarily from family or other people of similar well-being. Men had more information and training than women.

Conclusions

Social organization is important for leadership and community governance structures, agricultural support groups and social networks are important for learning. Formal leaders are more associated with innovation and adaptation, responsible for initiating changes, solving problems, disputes and making decisions within communities. Church was perceived to be influential to maintain wellbeing or to navigate change. Savings clubs for women had increased their access to financial capital, their participation in the community and household decision-making and increased their confidence, and their willingness to trial new innovations, like gardening schedules or increase crop yields and income.

Agency was perceived as people's ability to make their own choices or participate in household or community decisions which influence their ability to cope with or drive change to improve their wellbeing. *Wife must obey their husband* was frequently mentioned. But men were perceived to often make poor decisions concerning household money, alcoholism, etc.

The ability of people to maintain wellbeing is not built on assets alone. Learning and experimentation were influenced by access to information and the willingness to bear risks. These could in turn be influenced by the presence, absence or quality of relationships with external organizations. *Social and gender norms* shaped differences in women's and men's capacities, for example in assessing support and information, participating in organizations, learning and experimenting.

More livelihood opportunities may lead to a heavier labor burden for women in particular. Literature tends to emphasize the correlation between high livelihood diversity and high adaptive capacity but tends to overlook increases in labor burden and therefore this correlation does not necessarily equate to improved wellbeing.

An individual's capacity to access resources is strongly influenced by social norms built upon relationships of reciprocity and exchange (Adger 2003). Gender imbalances in education, physical mobility, agency and social standing meant that men were more able to access and utilize new sources of information. Women and youth in particular were less able to establish relations with external agencies to access new information and their capacities to innovate and adapt were restricted.

2. Walker, B., Holling, C.S., Carpenter, S.R. and Kinzig, A. 2003. Resilience, Adaptability and Transformability in Social-ecological systems. *Ecology and Society* 9: 5.

https://www.researchgate.net/publication/42764046_Resilience_Adaptability_and_Transformability_in_Social-Ecological_Systems

Research topic

The topic of this paper is the search for a better scientific basis for sustainable development, a new "sustainability science". It focuses on notions such as robustness, vulnerability and risk. Three complementary attributes are examined: resilience, adaptability and transformability, what they mean, how they interact and their implications for future wellbeing. The difficulty is that it deals with very imprecise concepts such as justice or wellbeing. The definitions can be too narrow and counterproductive. There is confusion about a resilience approach to contribute to sustainable development is mentioned. The article wants to provide an interpretation of how these concepts are reflected in the adaptive cycles of complex, multi-scalar SES. It does not refer to economic resilience, to gender or to social norms.

Methodology

What is new is the use of established theory of nonlinear stability (Levin 1999, Scheffer et al. 2001, Gunderson and Holling 2002, Berkes et al. 2003). This is used to examine regional development examples in different parts of the world (Everglades and Wisconsin Northern Highlands Lake District, USA, rangelands and agricultural production in southeastern Australia, semi-arid savanna in Zimbabwe, Water Kingdom in Sweden, Mae Ping valley in Thailand) based on literature review.

A major distinction is made between resilience, adaptability and transformability. Transformability relates to altering the nature of a system. The article introduces the concepts of adaptive cycles and cross-scale effects, which do not seem so relevant for HER+ research. It goes on to present four crucial aspects of resilience: Latitude - maximum amount a system can be changed before losing its ability to recover; Resistance: ease or difficulty of changing the system, how resistant it is to being changed; Precariousness: how close a current state of a system is to a limit; Panarchy: resilience of a system, which depends on influences from dynamics at scales above and below. Then more precise definitions of states, attractors and stability landscapes follow.

Relevant findings

The article includes many definitions of adaptability, collective capacity to manage resilience, transformability, etc.

Adaptability is the capacity of actors in a system to influence resilience. In a SES this amounts to the capacity of humans to manage resilience. A characteristic feature of complex adaptive systems is self-organization without intent (Levin 1998) cited on p. 4. The adaptability of the system is mainly a function of the social component – individuals and groups acting to manage the system, intentionally or unintentionally.

The collective capacity to manage resilience intentionally determines if they can succeed in crossing back into a desirable SES. Four ways to move away or come closer to the current system are presented: move thresholds away from or closer to the current state of the system; move the current state of the system away from or closer to the threshold; make the threshold more difficult or easier to reach; manage cross-scale interactions to avoid or generate loss of resilience at the largest and most socially catastrophic scales.

Transformability is the capacity to create a fundamentally new system when ecological, economic or social conditions make the existing system untenable, p. 4.

Resilience: more definitions about States, Attractors, and stability landscapes:

The state space of a system is defined by the state variables that constitute the system. The state of a system at any time is defined by their current values. p.4

A basin of attraction is a region in state space in which the system tends to remain.

The equilibrium state is defined as an “attractor” and the basin of attraction constitutes all initial conditions that will tend toward that equilibrium state.

We think of SES as moving about within a particular basin of attraction, rather than tending directly toward an attractor.

The various basins that a system may occupy, and the boundaries that separate them, are known as a ‘stability landscape’. (p. 4)

No SES can be understood by examining it at only one scale. The social component consists of groups of people organized at multiple levels with differing views as to whether some basins are desirable and others undesirable. The system is actually a subsystem of the whole panarchy (4th aspect of resilience) and resilience is influenced by what is happening in the panarchy at scales above and below the scale of interest.

More precise definitions of resilience, adaptability and transformability are proposed on p. 7.

Conclusions

Strategies for sustainability take many forms, no “one size fits all” approach to the future. The need to know the details of the local and regional context means a different approach to resource governance will be required for a sustainable future. Adaptive governance is a process of creating adaptability and transformability in SESs. Adaptive management has frequently failed. Because the stability landscape is constantly changing, the adaptive part of both governance and management is required in all phases of the adaptive cycle.

3. Smit, B., and Wandel, I., 2006. Adaptation, adaptive capacity and vulnerability. *Global environmental change* 16: 282-292

<https://www.sciencedirect.com/science/article/abs/pii/S0959378006000410>

Research topic

A review of the concept of adaptation in the context of adaptive capacity and vulnerability of human systems to global changes, especially climate change, with a special focus on practical applications of adaptation and adaptive strategies. The main purpose of participatory vulnerability assessments is to identify adaptation strategies that are feasible and practical in communities. Common elements of different forms of adaptation analysis are identified.

Adaptation in the context of human dimensions of global change usually refers to a process, action or outcome in a system (household, community, group, sector, region, country) in order for a system to better cope with manage or adjust to some changing conditions, stress, hazard, risk or opportunity.

Brooks 2003, p. 8 describes adaptation as “adjustments in a system’s behavior and characteristics that enhance its ability to cope with external stress”, Cited on p. 2

Smit et al. 2000, p. 225, in the climate context refer to adaptations as “adjustments in ecological-socio-economic systems in response to actual or expected climatic stimuli, their effects and impacts,” cited on p. 2.

Pielke (1998, p. 159) defines adaptations as the “adjustments in individual groups and institutional behavior in order to reduce society’s vulnerability to climate”, cited on p. 2.

Adaptations can be anticipatory or reactive, autonomous or planned.

This paper looks at applications to human systems and human-environment systems, including communities, households, groups, sectors, regions and countries.

Methodology

The focus includes the natural resource systems upon which societies depend, but the article does not review applications of physical or biological systems, but only in ecological systems, because the resilience concepts have been most developed there.

The article provides an overview of the concept of adaptation as it relates to adaptive capacity and vulnerability in the context of climate change. Several purposes of adaptation analysis are distinguished in climate change. Analytical approaches with practical purposes are reviewed to show that a bottom-up approach is needed based on empirical evidence on the ground.

Relevant findings

The origin of the term ‘adaptation’ can be found in natural sciences, particularly in evolutionary biology. The application of the term to human systems has been traced by anthropologist and cultural ecologist Julian Steward, who used “cultural adaptation” to describe the adjustment of “culture cores” to the natural environment through subsistence activities. Cited on p. 3

O’Brien and Holland 1992, p. 37 define the process of adaptation as “one by which groups of people add new and improved methods of coping with the environment to their cultural repertoire”, cited on p. 3.

Denevan 1983, p. 401 considers cultural adaptation as a “process of change in response to a change in the physical environment or a change in internal stimuli, such as demography, economics and organization”, broadening the range of stresses to which human systems adapt beyond biophysical stress”, cited on p. 3.

Cultural practices are equated with genetic characteristics in the natural sciences. A cultural practice is an adaptation only if it is developed to overcome stress. In more recent social science it is recognized that societies adapt to a range of stimuli including environmental stress. Cultures or societies which are able to respond to or cope with change quickly and easily are considered to have high “adaptability” or “capacity to adapt” (Denevan 1983), cited on p. 3. Adaptation is usually implicit in the political ecology field.

Four different purposes of climate change adaptation research are identified:

In a first group of studies, adaptations are considered to assess the degree to which they can moderate or reduce negative impacts of climate change, or realize positive effects, to avoid the danger. Analyses are undertaken at large scales, where statistical models are used to estimate impacts with and without adaptation. Adaptations are conventionally assumed or hypothetical and their effect is estimated. The focus is on the effect of the assumed adaptations.

In a second group of studies the focus is on specific adaptation options or measures for a particular system subject to climate change. To assess the relative merit or utility of alternative adaptations, in order to identify the best or better ones, selecting a suite of possible adaptations chosen from hypotheses, observations, modelling, extrapolation, analysis, key informants or deductive reasoning. Common variables to rank or rate the relative merit are benefit-cost, cost effectiveness and multiple-criteria procedures.

A third group of studies focuses on the relative adaptive capacity or vulnerability of countries, regions or communities and involves comparative evaluation or rating based on criteria, indices and variables selected by the researcher. Vulnerability is taken as the starting point rather than the end point. It is assumed to be measurable based on attributes or determinants selected a priori. The intent is to provide information for the targeting of adaptation initiatives, or the targeting of scarce resources. Used for rating or ranking analysis.

The fourth type of analysis is to contribute to practical adaptation initiatives. The focus is to document ways in which the system or community experiences changing conditions and the processes of decision-making in this system that may accommodate adaptations or provide a means of improving adaptive capacity. In the climate adaptation field, this body has several distinct features. It tends not to presume but seeks to identify empirically from the community. It takes a bottom-up approach in contrast to scenario-based, top-down approaches. Motivation: identify what can be done in a practical sense, in what way and by whom, in order to moderate the vulnerability to the conditions that are problematic for the community. Practical climate change adaptation initiatives are integrated with other programs and often aim to enhance adaptive capacity.

Conclusions

From adaptation analysis to practice. The community itself must identify the exposure and sensitivities and the determinants of adaptive capacity in the community. The methods require active involvement of stakeholders, information collection, integration of information from multiple sources and the engagement of decision makers.

Participatory vulnerability assessments allow for the recognition of multiple stimuli beyond those related to climate. Methodologies recognize the interaction of various exposures, sensitivities and adaptive capacities over time.

In the climate change field, adaptations can be considered as local or community-based adjustments to deal with changing conditions within the constraints of the broader economic-social-political arrangements. Where those constraints are particularly binding, adaptation may be considered as attempting to change those broad economic-social-political structures themselves.

- 4. Cramer, L., Förch, W., Mutie, J., Thompson, P. K., (2016) Connecting women, connecting men: how communities and organizations interact to strengthen adaptive capacity and food security in the face of climate change, gender, technology and development, 20:2, 169-199, <https://www.tandfonline.com/doi/pdf/10.1177/0971852416639771>**

Research topic

The article offers a cross-regional analysis from 15 sites across West Africa, East Africa and South Asia of local organizational landscapes as they relate to livelihoods and food security. Women tend to value local organizations more highly and thus appear to be less connected to external organizations than men. Most of the local organizations with food security as a stated objective focus on production, which can marginalize or alienate women.

Vulnerability to climate change and food insecurity is multidimensional and is affected by many factors, including location, livelihood system, socioeconomic status, gender, age and ethnicity.

The article analyzes how interventions to improve resilience are perceived by women and men and whether community and organizational priorities intersect. These are questions that have not yet been rigorously answered.

The following research questions are pursued:

1. Which levels and what types of organizations operate within the community and what is their value as perceived by men and women?
2. How are the components of food security addressed by these organizations perceived differently by men and women and what are the community identified gaps within the organizational portfolios?
3. What are the key disconnects between the gendered perceptions of self-identified organizational priorities and the identified priorities of men and women?

There seems to be little evidence of whether organizations are implementing interventions that effectively strengthen Adaptive Capacity (AC) of both men and women. However, there are some methodological issues which do not allow for coherent conclusions about this aspect, and which are highlighted by the authors themselves. There is also a problem with the comparability of different contexts, which remains unconvincing.

Methodology

The article presents a simple framework to integrate AC into the sustainable livelihood's framework (Scoones, 1998). Using gender-differentiated data from community-level participatory research and organizational level interviews from 15 sites across West- and East-Africa and South Asia, a cross regional analysis of local organizational landscapes is conducted as they relate to livelihoods and food security.

How *gender norms* affect the other elements of the framework at the community level is examined. A total of 675 women and 675 men participated in focus groups discussions with women and men separately. A total of 178 organizations were interviewed across all sites. AC is integrated into the framework to add a greater understanding of how communities can control their destinies and learn to change with their changing environments. The focus is on intersections of the sustainable livelihood's framework, SLF with AC and add the gender perspective to examine how women and men interact differently with their biophysical and institutional environments. The outline of data collection and presentation of results of the cross-regional analysis is linked to elements of the framework, with final recommendations for policy, programming and further research needs.

Relevant findings

Climate change will have broad impacts on food security and the consequences will be worse in those countries where high levels of food insecurity already exist. Most of the publications focus on the impact on production and availability, leaving gaps around access, utilization, and stability (Wheeler & von Braun, 2013).

Men and women do not produce food separately but play complementary roles in the production process (Doss 2014). Women more often spend income under their control on food and children's healthcare and education than men.

Encouraging collective action is one way of assisting women in improving the food security of their households. Women often rely on traditional social networks and social capital more than men due to their limited access to economic resources. Only working with women's groups can result in the ineffectiveness or failure of an intervention. Only 3 out of 87 articles included women, children and the elderly. Women often have lower AC than men.

Shift from the needs-based approach to an assets-based community development approach over the past decades. Rather than focusing on what communities lack, an effort is made to recognize what assets they possess.

From vulnerability to efforts of building AC. AC is defined as "the ability of systems, institutions, humans and other organisms to adjust to potential damage to take advantage of opportunities or to respond to consequences" IPCC, 2014 a. AC is closely linked to vulnerability and resilience. It emphasizes human agency and the capacity for forward-looking planned action as integral to building capacities to respond to change. The concept can be applied at various scales, from a national to an institutional, to a local, and a household level.

Gupta, Termeer, Klostermann, Meijerink, van den Brink, Jong et al. 2010 define and attempt to measure the AC of institutions using 6 dimensions: variety, learning capacity, room for autonomous, change, leadership, resources, and fair governance.

Recently there is more emphasis on processes, institutions, and broader contextual elements that influence AC. Varying influences on the capacity to adapt were found. AC cannot simply be built through interventions in the capitals. AC is a process which cannot be externally driven, but can be supported by external organizations, community-based organizations, and individuals.

AC is integrated into the sustainable livelihood's frameworks SLF (Scoones, 1998). The SLF framework considers how people's livelihood assets (human, natural, physical, social and financial capitals) are influenced by the vulnerability context and the structures and processes of their environment, and how these interacting factors affect livelihood strategies and outcomes. Political capital is added as a sixth asset.

Types of organizations and their perceived value

Women ranked organizations that focus on welfare, livelihoods, and social protection. Men focused on welfare, livelihoods, health sanitation and hygiene. Women valued savings and credit for women, capacity enhancement, education and savings and credit for men. Women identified more organizations working at least on one element of food security than men.

It is unclear however, whether the ways in which Community Based Organizations (CBOs) work is reinforcing traditional gender norms or challenging them and working toward gender equality. These results do not seem to be comparable between contexts and countries and are not very convincing.

Gender norms became more evident when results were broken down by food security component. Men named more organizations working on food availability, women more organizations working on food access and food utilization. Most organizations work on availability component, fewer are working on access. Food utilization received the least attention and was addressed mostly by external organizations. A strong focus on agricultural inputs was mentioned and little focus on capacity strengthening especially for women. Lack of attention to the elements of AC was highlighted.

Women would like more work being done on post-harvest storage and utilization. Women observed that they are not linked to extension services which entrenches existing *gender norms*.

Self-reported organizational priorities

External organizations reported fewer activities related to market access, income generation, post-harvest processing, access and utilization of related activities. Savings and credit was one of the most important thematic areas for women, but the area of lowest attention for organizations. There was no mention of building AC by organizations to respond to climate change. Few organizations state climate change adaption or mitigation as an objective at the locality.

Women tend to include local organizations more often into their top five than men. Men more often list national bodies in their list of top five. Women have less access to external resources than men and therefore less AC than men.

There seems to be a disconnect between community priorities and organizational objectives. In part this problem is related to how the data was collected.

Conclusions

Building the AC of communities to help them maintain and improve food security under a changing climate will require going beyond traditional development activities. There is a need to look beyond the technology transfer model and token nods for participation and develop ways of working with communities in a holistic manner while strengthening existing capacities, working with effective social networks, and targeting external investment in response to local priorities and needs. Women's self-help groups have strong social capital and could be utilized to increase food security and build community AC. A more holistic development approach also calls for more attention to the links between agriculture, food security, health and education.

Addressing climate change issues only through natural resource management (NRM) activities is also short of changing the possibilities for improving AC. NRM means a lost opportunity to critically reflect on and address other issues like power and political voice, governance and local leadership, as well as the role of external agency. Building community AC should be an explicit focus of development organizations. Men should not be left out of these considerations.

- 5. Jost, C., Kyazze, F., Naab, J., Neelormi, S., Kinyangi, J., Zougmore, R., Aggarwal, P., Bhatta, G., Chaudhury, M., Tapio-Bistrom, M-L., Nelson, C. & Kristjanson, P., (2016). Understanding gender dimensions of agriculture and climate change in smallholder farming communities, *Climate and Development*, 8:2, 133-144, <https://www.tandfonline.com/doi/full/10.1080/17565529.2015.1050978>**

Research topic

In Uganda, Ghana and Bangladesh the aim of this study was to deepen the understanding of gender in terms of the major topics of the global research program on Climate Change, Agriculture and Food Security CCAFS of CGIAR: climate smart agriculture, climate analogue approaches and climate and weather forecasting.

Topics selected by research leaders were engaging in climate-smart agricultural practices to sustainably increase productivity and resilience, reduce or remove greenhouse gases and enhance achievement of food security and development goals.

Policy and program-relevant results were obtained. Smallholders are changing agricultural practices due to observations of climatic and environmental change. Women appear to be less adaptive because of financial or resource constraints, because of male domination in receiving information and extension services and because available adaptation strategies tend to create higher labor loads for women. Social norms are not mentioned explicitly.

Methodology

The aim was to develop and test participatory tools for investigating the gender dimensions of agriculture and climate change. Tools were analyzed for ease and consistency of implementation, and the strengths of pilot test results for more gender equitable agricultural development strategies and solutions. Long term research sites 10 – 20 years were established in each region in the three countries mentioned in 2010, for exploration of diversity of socio-economic, ecological and production systems and solutions to climate change effects on agriculture and food security in these systems.

Focus groups were organized in three communities, one in each region with 15 – 20 individuals. No household participated in more than one focus group. The focus was on drivers for practicing climate smart agriculture (CSA), the organizations that promote it and the institutions that support it.

Key changes during the last 10 years of farming practices were discussed, including drivers of change, decision making related to change, constraints to participation, types and distribution of benefits, organizations involved in introducing or supporting change, the impact of change on overall wellbeing, income and food security. Key interviews with development workers were held to capture more information on the above issues.

Visits between villages were organized and male and female focus groups were formed. Key characteristics identified were drier climates in Ghana and Uganda, and higher soil salinity in Bangladesh. Village resource maps and guiding questions were used to explore perceptions regarding mobility and identify factors that may influence mobility and exchange of information pertaining to agriculture.

Access and use of daily weather and seasonal climate forecasts were analyzed by four focus groups for male and female adults and male and female young people; seasonal calendars were used to understand farming activities in relation to weather information.

Relevant findings

Farmers reported experiencing changes in rainfall duration and intensity, more erratic rainfall patterns and more prolonged droughts and difficulties in planning and making agricultural decisions. The emergence of new crop diseases and pests are associated with climate change. Drivers of changing agricultural practices were lower incomes, less household food availability, less access to resources for improved agricultural production, and more opportunities for training. Men were motivated by income opportunities to change their practices, women by increased food availability. Men have more access to resources like cash and are more enabled to take up new agricultural practices, so they are less vulnerable to the observed changes in weather patterns.

New crop management practices by women and men include intercropping, dry planting, earlier planting, adopting drought resistant varieties, and adjustments in the timing of weeding and harvesting. Some began rotating crops, integrating crops, livestock and trees, only a few reported improving seeds. More men participated in the construction of trenches for water management for banana and coffee fields.

Soil fertility enhancement with livestock manure and inorganic fertilizers were rare and if used this was only by men, due to the costs involved; men also adopted zero grazing practices, younger men adopted small scale irrigation practices for high value and labor-intensive crops such as tomatoes. Women reported using mechanical traps for pest control.

Age and sex are important difference in determining mobility, young men are most likely to travel; mobile phone owners travel less because they can access information about market opportunities and prices electronically. Factors limiting women's mobility included lack of funds to pay for transportation, poor infrastructure, security, health, household responsibilities and lack of permission from their spouses. Farmers travel by bicycle, but women would not ride bicycles so they must have cash for the bus fare.

Organization in production and marketing associations is facilitating adoption of improved agricultural practices. Farmers received weather information via SMS on their mobile phones by state institutions but found the information not credible. They did not understand the international weather symbols most often used in print media for daily forecasts and found seasonal forecasts difficult to understand. Few had ever received a seasonal forecast. Women did not understand them and the implications for agricultural decision making while some men were able to do so.

However, forecasts broadcasted by radio were found unreliable as they related to a large geographic area and not their village specifically. Women and men received climate information from trusted elders within their family. Men prefer to hear information on the radio; women prefer megaphones, letters, village leaders, farmers groups, school children, religious and social gatherings and print media, but only useful when in the local language and more location specific.

The article includes more detailed descriptions of the three countries.

Conclusions

The climate analogue approach (identifying places resembling your future climate to identify potential adaptations) is a promising tool for increasing farmer-to-farmer learning, where a high degree of climatic variability means that analogue villages that have successfully adopted new CSA practices exist nearby. Institutional issues related to forecast production limit their credibility and salience, particularly in terms of women's ability to access and understand them. The participatory tools used in this study provided some insights into women's adaptive capacity in the villages studied, but not to the depth necessary to address women's specific vulnerabilities in CSA programs. Further research is necessary to move the discourse related to gender and climate change beyond the conceptualization of women as a homogeneously vulnerable group in CSA programs.

There is potential for spreading knowledge of adaptation strategies via the use of a climate analogue approach, particularly at a local level but not only based on environmental characteristics. Social and cultural contexts matter a lot. Examine in more detail how different and changing social and cultural contexts in climate analogue locations affect peoples' and communities' capacities to adapt. The concept of positive deviance can be used to identify innovative farmers' testing adaptation options within a potential analogue community (Biggs 2008).

Indigenous knowledge is the most available and considered the most reliable source of weather and climate information in all three sites. Indigenous forecasters are an entry point for the joint production and distribution of seasonal climate forecasts. Elders of both genders produce and disseminate this information in the three study communities. However, the extent to which indigenous knowledge is more accurate than scientific forecasts is yet to be tested in a systematic manner.

Suggestions to improve the credibility and salience of seasonal forecasts have included downscaling, use of local languages, inclusion of recommendations on how the information can be used, facilitating access to the resources and inputs that farmers need in order to take advantage of agro-meteorological information and participatory approaches to dissemination at the local level. Women's access to information appears to be limited by their lack of technology ownership.

“ Indigenous knowledge is the most available and considered the most reliable source of weather and climate information in all three sites.

6. Rao, N., Lawson, E.T., Raditloaneng, W.N., Solomon, D. & Angula, M.N. (2019). *Gendered vulnerabilities to climate change: insights from the semi-arid regions of Africa and Asia*
<https://www.tandfonline.com/doi/full/10.1080/17565529.2017.1372266>

Research topic

Nearly all policies aimed at developing and strengthening the adaptive capacity of local communities fail to recognize the gendered nature of everyday realities and experiences. Women are portrayed as vulnerable, weak, poor and socially isolated, and men are invisible from much of this discourse; their absence from the locality is only seen as enhancing women's vulnerability to risks and stresses.

What emerges from the field is the "multiplicity, intersectionality and everyday nature of the risks and stresses that characterize life for poor people" Terry 2011. The IPCC fifth assessment report has acknowledged the overlapping and intersecting nature of risks – geophysical, agro-ecological and socio-economic- when it states with very high confidence that differences in vulnerability and exposure arising from non-climatic factors shape differential risks to climate change.

The aim of this paper is to develop a broad-based understanding of gendered vulnerability as emerging from poverty and social discrimination and socio-cultural practices in different political, geographical and historical settings, apart from climatic variability and environmental/natural risks.

Research questions

How do differences in household structures and conjugal relations, the division of labor and rights and responsibilities embedded therein, shape adaptation?

What are the trade-offs involved in the choices people make – between short-term coping and longer-term adaptation, between nurturing social relations of reciprocity and interdependence and seeking individual welfare?

Access to resources is important, but how these link to social roles, *norms*, *values* and cultural identities in different contexts?

Methodology

This paper builds on a regional diagnostic analysis of vulnerability and adaptation to climate change in semi-arid regions (SARs) across Africa and Asia conducted by the ASSAR (Adaptation at Scale in Semi-Arid Regions) project along with preliminary field observations to explore some of these puzzles. It includes regional comparisons and very interesting findings and conclusions with several references to *social and gender norms* and how they limit women's options to react to climate change. Social relations and their gendered symbolic meanings are highlighted as very important for adaptive capacity.

Relevant findings

The 'virtuous women and negligent men' discourse, with its focus on women's vulnerability and virtuousness, deflects attention away from the real inequalities in decision making and resource access, as well as institutional norms that exacerbate women's exclusion. There exists a risk of instrumentalizing women's labor for achieving development goals.

A more nuanced understanding of intersectionality is needed in terms of multiply determined, simultaneous and interlinked experiences of power and inequality, privilege, and oppression, across scales, from the micro to macro.

Stark differences were found in vulnerabilities and coping strategies of households with seasonally irrigated land, those dependent entirely on rainfed farming and landless laborers. Workloads increased most for women in rainfed farming

households due to fluctuating crop yields, longer distances to travel for fuel, fodder and water for their livestock, with adverse health consequences.

The landless moved to non-farming work outside the village and in households with irrigation, there were some shifts in work patterns due to the adoption of short duration crops.

If policies focused only on the lack of assets, like landholding, the most vulnerable households, the rainfed farming households and women within them, would be missed out. Gender here works not on its own, but in interaction with the nature of farming practiced.

Bryceson 2013 documents the range of relationships and partnerships – economic and sexual – that are being formed and negotiated between men and women in a context of stress and competition for scarce resources in Tanzanian gold-mining settlements and similar in Northern Kenya.

Male invisibility from the household economy can have several negative effects – from male withdrawal to expressions of violent masculinities. Non-recognition and even vilification for neglecting their responsibilities can aggravate male sensitivities.

What is needed is to support and facilitate cooperation within households and communities, rather than targeting either men or women, often placed in opposition to each other.

Resources have different material and symbolic meanings and access mechanisms. Availability and access to water and pasture lands are equally important as cultivable lands.

Men and women with social support systems and some assets, are potentially better able to manage climatic and livelihood uncertainties than those without.

It is important to consider resource access, not only at the individual household level as a personal asset but rather how it is distributed and governed across scales from the local community to the region and even the nation-state.

Men and women have complementary roles shaped by *cultural values and social norms* and shaping adaptation responses to climatic and other stresses.

“ Workloads increased the most for women in rainfed farming households due to fluctuating crop yields, longer distances to travel for fuel, fodder and water for their livestock, with adverse health consequences

Conclusions

Key lessons for policy, practice and research are to: exercise caution while generalizing, and to recognize differences in strategies and engage with them in nuanced and disaggregated ways; move beyond counting numbers to unpacking relations of power, of inclusion and exclusion in decision-making processes and challenging cultural beliefs that have denied equal opportunities and rights to differently positioned people, especially those at the bottom of economic and social hierarchies; and engage with social complexity in terms of the intersections of gender, race, ethnicity, or class in specific local settings and explore in which ways men and women together or separately in their different and changing roles, cope and adapt to changing circumstances, while also shaping the outcomes of external interventions.

Gendered vulnerabilities and adaptive responses: regional comparisons

There are different impacts of change for women and men, belonging to different wealth, age and status groups, potentially enhancing the risks and vulnerabilities they face. In the process, gender relations and the organization of social reproduction are likely to change, though the direction of change is not necessarily predictable. Women's position may be strengthened, or gender inequalities may get further intensified.

It is important to document differences in perceptions and question dominant narratives about men's and women's roles in the economy.

“ There are different impacts of change for women and men. Women's position may be strengthened, or gender inequalities may get further intensified.

Water security and drought

Migration emerges as an important adaptive strategy in the face of climate and other livelihood risks and uncertainties. Migration patterns are gendered. In many cases women take on extra workloads in their efforts to cope with male migration, adopting a host of diversification strategies including trade, to provide food for the household.

Married women may be excluded from profitable activities due to local taboos as well as domestic responsibilities. When pressures become too burdensome, women opt out of marital relationships, retaining responsibility only for themselves and their children. In some regions there is a rise in female-headed households.

The individual vs. the collective: *norms, values*, and the temporality of choice

Empirical studies exploring cultural differences on adaptation are rare. Culturally specific views of the good life as well as historical processes have led two ethnic groups living in the same Sahelian village to take different livelihood paths in a context of drought and economic pressure.

A key justification for women's land claims relate to their significant labor contributions to farming, together with their insecurity in terms of returns or benefits. Land tenure security is viewed as crucial for both pastoral and agrarian livelihoods.

Assets are not just material resources in the hands of individuals and communities, but are imbued with symbolic meanings around security, status, and respectability. They are embedded in social relations and contribute to perceptions of wellbeing. The meanings and values of assets change with shifts in development paradigms whether this is through state control or markets. The gendered meanings of assets need to be understood in each context. Universal policies around assets are not always practicable, nor will they necessarily enable progress towards gender equitable or indeed transformative adaptation.

Collective action, voice, and decision-making: what is visible to policy?

Focus on the public and the productive in policy interventions tend to invisibilize adaptive responses that lie in the private or reproductive realm (using cheaper foods, skipping meals) and this excludes women from the more strategic decisions relating to the allocation of both household and community resources.

Collective action does have empowerment effects, beyond the immediate purposes of resource conservation and management and can help challenge oppressive gender and wider social relations.

Gains are not only material, but also personal and relational, strengthening skills and dignity and gaining voice within groups and networks.

Farmers' groups, community networks, credit groups and village councils are all seen to have a positive influence on adaptation. Examples of playing gender roles rather than challenging social rules and norms were important to gain leadership positions for women.

It is important to recognize cooperation and conflict between men and women in the household and across institutions, and the continuous negotiation of power relations, through open struggles or more covert, hidden forms of influence. Negotiations are

reflected in private adjustments as well as in more public forms of claim-making and collective action. Women's strategies span the two domains, yet policy attention largely targets the more public and visible, rather than alleviating women's everyday burdens.

A landscape approach to adaptation strategies, rather than either group or resource-based interventions (Ahmed et al. 2016; Batterbury 2001), addresses a host of interconnected issues in terms of barriers and enablers within a particular landscape. In times of crisis, social relations trump all over resources, material, and non-material.

The reorganization of domestic groups and the rise in the numbers of female headed households, growing resource conflicts around both water and land use and management, and their gendered subtexts, particularly with the monetization and commoditization of these resources, are all significant. New forms of diversification and collective action are emerging, especially by women.

7. Abdul-Razak, M., & Kruse, S. (2017). The adaptive capacity of smallholder farmers to climate change in the Northern Region of Ghana. *Climate Risk Management*, 17, 104-122.

<https://doi.org/10.1016/j.crm.2017.06.001>

Research Topic

Abdul-Razak & Kruse (2017) sought to characterize and assess the adaptive capacity of smallholders in northern Ghana, then identify determinants of high and low capacity.

Methodology

Drawing on literature review, the authors deductively developed an indicator framework for adaptive capacity and verified it with local expert input (notably, not with farmers). The framework included six dimensions, weighted in the following order: economic resources (income diversity, remittances, credit access), awareness and training (including literacy, climate information, extension, farming experiences), knowledge of agricultural technologies, infrastructure (including roads, irrigation, landholding), social capital (including access to family labor, participation in various groups), and institutions (including land tenure, subsidies, disaster relief). Based on this framework, the authors conducted a survey with 80 smallholders (non-randomized) in two communities in northern Ghana and used the data to develop adaptive capacity scores for each farmer.

Relevant findings

Abdul-Razak & Kruse (2017)'s results showed that men had greater adaptive capacity overall and across all six dimensions they considered in their scoring process. However, social capital and institutional factors were similar between genders. Their findings indicated that economic resources, awareness and training, and technological capacities were the dominant factors limiting women's adaptive capacity. After categorizing farmers according to their adaptive capacity (from "very high" to "very low"), the authors found that no respondents of either gender had "very high" adaptive capacity, and no women fell into the "high" category. Rather, twice as many women as men (78% of women) had "low" adaptive capacity.

Conclusions

Abdul-Razak & Kruse (2017) concluded that the key drivers of the gender gap in adaptive capacity related to gender disparities in access to education, diversified income sources, credit, technical knowledge of farming practices, climate information, and extension services. They advocated for expanded training and education programs.

Reviewer comment

Relevant sections: income generation options, information/extension, education, credit.

8. Andrews, M. (2019). A case study of the Southern African Rural Women's Assembly: "We can bend the stick." *Agenda*, 33(1), 48-58. <https://doi.org/10.1080/10130950.2019.1598275>

Research topic

Andrews (2019) explored the Rural Women's Assembly in Southern Africa as an example of feminist leadership formation and movement building.

Methodology

Andrews (2019) used the grassroots Rural Women's Assembly (RWA) in Southern Africa as a case study illustrating rural women's leadership and challenging of cultural norms related to, e.g., subservience. They drew on participant observation and document analysis, primarily, to share the story of RWA's founding and establishment of key group strategies and approaches.

Relevant findings

Andrews (2019) discussed the establishment and evolution of the RWA as a group that "fights injustice, inequality, poverty and sees itself as a non-patriarchal, anti-capitalist, anti-racist movement that is inclusive and is opposed to all forms of prejudice, especially xenophobia and homophobia" (p. 54). In the process, the author briefly discussed challenges that rural women in this context faced, including social norms related to ownership/inheritance, control, and access to land that limit women's autonomy and involvement in agricultural decision-making; those limiting women's engagement in local governance; and those prescribing domestic and caregiving activities to women.

Conclusions

The author concluded that RWA had intentionally built "a methodology and a praxis that challenged the dominant values of a male-centred, hierarchical, directive-oriented, centralised organisational form" (p. 56). The group pursued these goals by supporting women in taking leadership roles, involving women in governance intended to benefit them, using collective reflection and discussion to build "a common and grounded analysis of the power relations that shape rural women's lives" (p. 56), and underscoring the importance of women-only organizations.

Reviewer comment

This may not be relevant due to lack of primary empirical data. It presents observational data. Relevant sections on land, decision-making; touches on division of labor.

- 9. Antwi-Agyei, P., & Nyantakyi-Frimpong, H. (2021). Evidence of climate change coping and adaptation practices by smallholder farmers in northern Ghana. *Sustainability (Switzerland)*, 13(3), 1-18. <https://doi.org/10.3390/su13031308v>**

Research topic

Antwi-Agyei & Nyantakyi-Frimpong (2021) looked at relationships between short-term coping strategies and long-term adaptations among smallholder farmers in three districts of northern Ghana.

Methodology

The authors surveyed 555 household heads across six communities to assess sociodemographics, perceptions of climate change, access to climate information services, and short- and long-term measures employed in response to climate change. They used descriptive statistics to evaluate the relative importance of adaptations. Follow-up focus group discussions were also employed.

Relevant findings

The top reported adaptations were crop diversification, planting drought tolerant crop varieties, and pursuit of non-farm livelihoods—only the latter was significant different between genders (practiced more by women). Among coping strategies, men were significantly more likely to report engaging in wage labor and women more likely to engage in petty trade and reduced food consumption (as primary caregivers). Women mentioned more constraints to adaptation than men, on average, including access to markets (tied to their cultural responsibility for marketing products), lack of irrigation infrastructure (tied to men's control over these facilities), secure access to fertile land (tied to norms around inheritance and exacerbated by men's emigration), and lack of labor (tied to prioritization of the husband's fields). Access to inputs also emerged as a constraint in Focus Group Discussions.

Conclusions

The authors concluded that key intersecting barriers to adaptation were lack of access to credit, markets, land, inputs, irrigation facilities and limited institutional support. They emphasized the need to address gender disparities, reform social norms, and implement "gender-based climate change interventions" (p. 15), given evidence of gender-based differences in barriers to adaptation.

Reviewer comment

Relevant sections: access to land, access to resources, decision-making

10. Asante, F., Guodaar, L., & Arimiyaw, S. (2021). Climate change and variability awareness and livelihood adaptive strategies among smallholder farmers in semi-arid northern Ghana. *Environmental Development*, 39(November 2020), 100629. <https://doi.org/10.1016/j.envdev.2021.100629>

Research topic

Asante et al. (2021) evaluated Ghanaian farmers' perceptions of climate change, impacts on farming systems, and adaptive strategies (especially livelihood strategies) employed in response.

Methodology

The authors surveyed 121 farmers (50% female) and conducted interviews and focus group discussions. They analyzed the qualitative data (although it is not much referenced in the text) and used regression models to identify predictors of adaptation.

Relevant findings

The authors found widespread perceptions of climate change and variability, with observations of climate change more common among men and more educated farmers. In terms of adaptations, the types of off-farm activities that farmers turned to were found to be highly gendered. Women in this study appeared to have more livelihood options than men and engaged in various forms of petty trade, tailoring, hairdressing, and small business ownership, relative to men's engagement only in charcoal production, hunting and carpentry. The authors attributed this division of livelihoods to differing physical capabilities. Older farmers, men, and those with access to credit were most likely to practice at least one adaptation. The authors indicated that the gender gap in adaptation appeared because men were more engaged in farming (which is 'masculine in nature') and women more engaged in off-farm work, although they also stated that women often have fewer financial opportunities and less adaptive capacity. Other constraints to adaptation documented in the study appeared to have gender implications, though the authors did not explicitly address them: changing the location of farmlands was constrained by land tenure, finance, and distance, while use of irrigation technologies was constrained by access to capital. Gender norms were indirectly addressed for the most part, and the authors did not offer a clear basis for their explanations of gendered patterns in adaptation.

Conclusions

The authors concluded that farmers in the study area widely perceived harmful climate change impacts. Non-farm activities were largely preferred as adaptation strategies and were highly gendered. The authors advocated for more location-specific adaptation policies that account for sociodemographic variability, integration of indigenous knowledge, improved access to credit and education for farmers.

Reviewer comment

Relevant sections: access to credit. Minimal engagement with gender and rather unclear explanations of their findings.

11. Autio, A., Johansson, T., Motaroki, L., Minoia, P., & Pellikka, P. (2021). Constraints for adopting climate-smart agricultural practices among smallholder farmers in Southeast Kenya. *Agricultural Systems*, 194(January), 103284. <https://doi.org/10.1016/j.agsy.2021.103284>

Research topic

Autio et al. (2021) evaluated biophysical and socioeconomic constraints to adoption of climate-smart agriculture (CSA) practices in southeastern Kenya.

Methodology

The authors used rapid and participatory rural appraisal tools, including 65 household surveys of farming activities and land tenure systems, gender-disaggregated farmer workshops with 102 participants (50 women), key informant interviews with

government representatives and others, and transect walks. Workshops generated cropping and climate calendars, resource maps, and pairwise rankings of CSA practices according to awareness and use.

Relevant findings

First, the authors noted that women and children in this context were especially affected by drought due to water collection responsibilities. They found that while state actors believed that the core obstacle to adoption of CSA was farmers' lack of awareness, farmers themselves reported a range of constraints to use of CSA, notably access to markets, land tenure, and lack of resources. The authors discussed several gender-related implications of these barriers. First, they noted that decision-making in these farming systems was largely controlled by men, who also have more secure land tenure under traditional systems. These factors limited women's ability both to adopt CSA and, potentially, access credit (as this is tied to land tenure): "women's marginalization in TTC from decision-making processes, as described in the household roles, lack of property rights, and limited access to credit and other production inputs, hinder them from investing in CSA technologies and practices to achieve their optimal production potential" (p. 11). Gender norms in these farming systems also established a gendered division of farm labor, with crops for home consumption and less physically demanding tasks typically assigned to women. The authors stated that this division meant that adaptive measures that improved agriculture's commercial potential were less beneficial to women. Access to information and farmer groups was considered equal or favored women so did not emerge as a barrier. Still, there were some gender-based differences evident in *which* CSA practices men and women were aware of and used – men appeared to have more awareness of practices based around organic inputs (composting, mulching, green manure), while women appeared to use more low-input practices such as intercropping, fallowing, and organic agriculture.

Conclusions

The authors concluded that differences in farmers' biophysical environments and sociodemographics have contributed to heterogeneity in adoption. They called for more locally adapted and gender-responsive policies and programming for both extension services and development interventions.

Reviewer comment

Relevant sections: access to land, access to capital/credit, decision-making, domestic responsibilities

12. Awazi, N. P., Tchamba, M. N., & Avana, T. M. L. (2019). Climate change resiliency choices of small-scale farmers in Cameroon: determinants and policy implications. *Journal of Environmental Management*, 250(May), 109560. <https://doi.org/10.1016/j.jenvman.2019.109560>

Research topic

Awazi et al. (2019) examined Cameroonian farmers' adaptation measures on-farm (including agroforestry) and off-farm and sought to understand determinants of adaptive behaviors.

Methodology

The authors organized focus group discussions, key informant interviews with local leaders and government officials, and household surveys with 300 household heads across ten villages in two districts of northwestern Cameroon. Surveys captured socio-demographic data, perceptions of climate variability, and on-farm, off-farm, and agroforestry-related adaptive measures, which were then analyzed largely using regression models to identify predictors of adaptation.

Relevant findings

Awazi et al. (2019) found that gender (seemingly the gender of the respondent, although the variable is not explained) was *not* a significant predictor of adaptation overall. However, a number of factors typically correlated with gender were found to be associated with adaptation. The age of the household head, number of farms (fields?), farm size, household income, farm experience, and exposure to extreme weather events were positively correlated with adoption of on-farm and agroforestry-based adaptation measures. Access to weather information, extension services, credit, markets, as well as farmer group membership, were all positive correlated with use of on-farm, off-farm, and agroforestry-based adaptations. The authors did not address gender norms in this context, although their findings around access to information, extension, and credit may have implications for gender.

Conclusions

The authors concluded by restating the variables they found to be significant in predicting adaptation, and advocated for policies to support land ownership, farmer groups, access to weather and farming information, access to markets.

Reviewer comment

Relevant sections: access to information, credit. Gender norms are not directly addressed.

13. Cáceres-Arteaga, N., Maria, K., & Lane, D. (2020). Agroecological Practices as a Climate Change Adaptation Mechanism in Four Highland Communities in Ecuador. *Journal of Latin American Geography*, 19(3), 47-73. <https://doi.org/10.1353/lag.2020.0071>

Research topic

Cáceres-Arteaga et al. (2020) sought to evaluate whether implementation of agroecology as a climate change adaptation mechanism impacted socioeconomic conditions in farming communities in north-central Ecuador.

Methodology

The authors first conducted fifteen stakeholder interviews with elected leaders, technical personnel, social organization leaders, and farmers to assess perceptions of climate change and its impacts, challenges in adaptation, and the potential for agroecology to assist. The authors then conducted a survey with 119 farmers who participated in a past agroecology project, covering the same topics in addition to labor and demographic data.

Relevant findings

Cáceres-Arteaga et al. (2020) found that, in their study context, women faced heavy labor burdens, reporting higher daily engagement in both productive and reproductive activities than men. However, women worked outside the household only in agriculture, while men reported engaging in a wider set of income-generating activities. In examining impacts of the agroecology project, the authors found that farmers perceived agroecology to be a useful means of adapting to climate change. Women in particular derived numerous benefits, especially as the project challenged social norms that relegated reproductive work to women and left them limited control over income. Weekly 'agroecology fairs' enabled women to enter the public space and proved a boon in terms of women's income diversification, greater mobility, and increased social capital, autonomy (including increased financial control), and self-esteem. These activities also empowered women to learn and share ancestral knowledge of crops and wild plants more readily and to improve their household diets. However, women's experiences differed according to their marital status. Married women reported sharing land tenancy with their spouses, depending on their spouse as the primary source of household income, and having less control over agroecology-derived income; only single, divorced, or married women reported having significant power within their household.

Conclusions

The authors concluded that agroecology not only supported climate change adaptation but challenged gender inequalities. Women's adaptive capacity increased through agroecology project, as it improved their "access to education and training, the ability to generate income, and participation in community organizations" (p. 67).

Reviewer comment

Relevant sections: income generation options, behavioral norms (mobility, public vs. Private space), education, social capital, labor.

14. Erwin, A., Ma, Z., Popovici, R., Salas O'Brien, E. P., Zanotti, L., Zeballos Zeballos, E., Bauchet, J., Ramirez Calderón, N., & Arce Larrea, G. R. (2021). Intersectionality shapes adaptation to social-ecological change. *World Development*, 138, 105282. <https://doi.org/10.1016/j.worlddev.2020.105282>

Research topic

Erwin et al. (2021) sought to understand perceptions of socio-environmental change and private (individual/household) and public (collective) adaptation strategies in Peru. They also explored how intersectionality impacted adaptive capacity.

Methodology

The authors used participant observation and conducted semi-structured interviews with agricultural system actors in Peru (58 women and 128 men, including farmers, laborers, and irrigators). Interviews addressed perceptions of and responses to socio-environmental change, both as individuals or households and as communities. In the process, the authors assessed intersecting aspects of identity including gender, age, land ownership, education, and language.

Relevant findings

Erwin et al. (2021) found that agricultural system actors observed a wide range of socio-environmental changes, including climate change and decreasing water availability, worsening animal and plant health, changing cultural practices, increased migration, and widespread agricultural commercialization. Within the private space, the authors found that respondents adapted to change through income diversification and labor migration. Among 'unskilled' workers (those lacking formal education, typically), they found men had more income generation options (in mining and construction, e.g.) than women, who mostly worked as agricultural day laborers: "intersecting vulnerabilities associated with being a woman and 'unskilled' limited interviewees' power to access other income diversification opportunities, thus limiting their individual, and at times their household's, capacity to adapt to social-ecological change" (p. 8). Within the public space, agricultural system actors adapted to change primarily by engaging in meetings, discussions, and planning processes. Gender also emerged as a factor in the public adaptation space, as women "rarely voiced their opinions during meetings" (p. 9) due to social norms and language barriers. Meetings were held in Spanish, which older women in particular were not always proficient in due to lack of educational opportunities, and "were often designed to meet the needs of Spanish-speaking men who own land" (p. 9); "only community members who owned land could vote on water management rules and projects as part of an irrigation commission" (p. 10). The authors emphasized that "structural power in public spaces is not only legitimized through formal rules such as landownership; but it is also reinforced by who historically and continually interacts with institutions, the ways formal institutions and the state shape indigenous systems, and the associated social norms related to language and gender" (p. 10). The authors also found that older respondents were found less likely to migrate out or diversify incomes beyond farming or pastoralism. Finally, they documented that migrants who lacked resources to purchase land often lived in informal settlements and lacked secure water access.

Conclusions

Erwin et al. (2021) concluded that both public and private adaptations were shaped by agricultural system actors' intersectional identities. The factors most associated with constraints on adaptation in both the public and private space included gender, land ownership, and education. Thus, women with limited access to land and less formal education (and therefore weaker Spanish skills) faced many interacting constraints.

Reviewer comment

Relevant sections (many): education, land, behavioral norms, income generation options, poverty, intersectionality.

15. Guodaar, L., & Asante, F. (2018). Using a factor analysis to understand climate adaptation barriers impeding smallholder tomato farmers in the Offinso North District, Ghana. *Cogent Food and Agriculture*, 4(1). <https://doi.org/10.1080/23311932.2018.1504507>

Research topic

Guodaar & Asante (2018) assessed barriers to climate adaptation among smallholder tomato farmers in a district in southern Ghana.

Methodology

The authors conducted surveys with 378 tomato growers, collecting social and economic data and farmers' self-reported adaptive measures and barriers to adaptation. They then used factor analysis to quantitatively evaluate the importance of different barriers to adaptation and employed regression to identify socioeconomic factors associated with these constraints.

Relevant findings

The principal factors identified as barriers to adaptation were personal barriers (finance and knowledge), institutional and labor barriers (primarily extension limitations and cost of labor), irrigation technology barriers, credit and inputs access barriers, cost of land barriers, processing facility access barriers, and lack of political will. They found older farmers, men, and married farmers were more likely to report barriers to adaptation. The fact that men were more likely to report adaptation barriers is attributed to their responsibilities to support the household and their dominant role in tomato production systems, which likely exposes them to more problems, paired with women's relatively diversified sources of income. Additionally, the authors argued that married farmers faced greater financial pressures to support their household. Women in this system reported better access to credit than men, reportedly due to preferential treatment by lenders (due to women's higher repayment rates). The authors' regression model (in which the reference category was farmers reporting no constraints to adaptation) does not contribute to understanding differential barriers to adaptation, which limits its relevance for this study.

Conclusions

The authors concluded that farmers are adapting both on- and off-farm, although some individuals are unable to practice any adaptations. They advocated for financial support to farmers, and improved extension services

Reviewer comment

Despite gender-disaggregated data, there is no explicit discussion of norms and no breakdown of barriers by gender.

16. Hossain, K. M., & Zaman, F. (2018a). Unravelling coastal people's adaptation to salinity: evidence from Bangladesh. *International Journal of Environment and Sustainable Development*, 17(1), 70-92. <https://doi.org/10.1504/IJESD.2018.089278>

Research topic

Hossain & Zaman (2018a) focused on individuals' adaptive responses to saltwater intrusion in Bangladesh, with special attention to gender.

Methodology

The authors conducted surveys with 318 landless or minimal-landholding people (195 women, 123 men) in two coastal communities in Bangladesh. In the surveys, they captured social and economic data related to capitals and assets, then invited individuals to list and prioritize their adaptive strategies using Likert rating scales.

Relevant findings

They found that adaptive strategies that required financial inputs (e.g., introduction of crop varieties, changing land uses, well installation) were generally adopted by men, along with livelihood diversification and migration. Women, lacking capital, adopted a smaller range of practices that relied more on their own labor and human capital (e.g., water collection, water harvesting, cultivation of fish fry in reservoirs). Both men and women relied on social capital, but women faced norm-related barriers to engaging with and receiving trainings from NGOs. Women were generally involved in subsistence activities at the homestead, while men engaged in other livelihood activities (mainly NTFP). The physical isolation of these communities meant most women selling produce were forced to take low prices from local buyers, and their movement (as well as men's) was constrained. Women also faced disproportionate natural capital constraints in their limited access to land.

Conclusions

The authors argued that adaptive capacity in this context was constrained by resource entitlements, for women especially, who faced a triple burden of responsibilities linked to childcare, household nutrition, and marketing produce. They advocated for investing in women to support both gender equality and household wellbeing.

Reviewer comment

Relevant sections: access to finance, access to household resources, behavioral norms (mobility, engaging with NGOs), income generation options.

17. Hossain, M. A. R., Ahmed, M., Ojea, E., & Fernandes, J. A. (2018b). Impacts and responses to environmental change in coastal livelihoods of south-west Bangladesh. *Science of the Total Environment*, 637–638, 954–970. <https://doi.org/10.1016/j.scitotenv.2018.04.328>

Research topic

Hossain et al. (2018b) sought to understand the perceived impacts of climate change and human activities on aquatic systems in Bangladesh, as well as adaptive strategies employed by people involved in related livelihood systems.

Methodology

The authors first conducted a literature review and expert consultations to identify key topics of concern, then organized six mixed-gender focus group discussions with fishers, fish farmers, and retired fishers living in the cyclone-prone southwestern Bangladesh floodplains. These were paired with a few key informant interviews and observation.

Relevant findings

The authors found that, in general, adaptive responses in this region are oriented around “decreasing livelihood vulnerability, sustainable fisheries management, livelihood diversification, fisher mobility, planned adaptation, access to markets, spatial management, economic stability, and scientific knowledge” (p. 963). In this context, although women occupied a relatively limited role in the fish sector overall, they played a dominant role (80% of the workforce) in fish processing and drying. The authors stated that limited availability of fresh drinking water placed added labor burdens on women and girls who customarily collect it for the household. Additionally, the conversion of rice paddies to shrimp production had drastically reduced the availability of rice straw for cooking fuel, adding to women’s resource collection burdens. Women contributed not only to household reproductive labor but provided hired labor in fields and fisheries, where they reported receiving substantially lower (20% less, at least) casual wages than men. However, they lacked “voice to break through the long run discrimination” (p. 963). Women also faced institutional barriers to diversifying their incomes—specifically a lack of institutional credit support from government organizations for women’s traditional livelihoods. The mixed-gender focus group approach likely limited further insights in gender-differentiated experiences.

Conclusions

The authors concluded that communities in this region were reshaping production systems in response to changes, but that new models were not necessarily sustainable. They advocated for greater support to communities overall through training and education and support for alternative livelihoods, and especially to women, including minimum wage regulations improved credit offerings to support livelihood diversification.

Reviewer comment

Relevant sections: labor/time burdens, income generation options.

18. Jellason, N. P., Conway, J. S., & Baines, R. N. (2020). Exploring smallholders’ cultural beliefs and their implication for adaptation to climate change in North-Western Nigeria. *The Social Science Journal*. <https://doi.org/10.1080/03623319.2020.1774720>

Research topic

Jellason et al. (2020) examined relationships between religious (cultural) beliefs and adaptation among farmers in Nigeria.

Methodology

The authors held six gender- and age-disaggregated focus group discussions involving 61 farmers in two predominantly Islamic communities in northwestern Nigeria, as well as several key informant interviews. Focus groups were designed to gather an

understanding of dominant farming practices, perceptions of weather and climate variability, impacts on households and livelihoods, and adaptive responses—including decision-making around adaptation and the viability of strategies. Data were coded and qualitatively analyzed for themes.

Relevant findings

The authors found that older women and female household heads engaged in crop production in this context, but seldom played a role in decision-making in male-headed households. Younger women were not generally involved in farming “due to cultural reasons” (p. 19). However, women who farmed perceived the effects of climate change on yields. The authors documented several on- and non-farm adaptations (e.g., planting of early maturing varieties, dry planting, tree planting, livelihood diversification, and asset sales), but religious adaptations were the predominant strategies across focus groups. They tied religious adaptations to a lack of education. However, the authors did not discuss in-depth the role of gender or social norms in relation to adaptation, and did not have a robust household- or individual-level dataset that would allow them to speak to drivers of farmers’ decisions.

Conclusions

The authors concluded that the reliance on religious adaptations was likely a form of maladaptation. Thus, they argued that adaptation policies and strategies should respond to and accommodate the centrality of religion and sociocultural adaptation, but also improve farming households’ capacity to engage in more practical adaptations alongside prayer.

Reviewer comment

Relevant sections: divisions of labor, intersectionality.

19. Jiri, O., Mafongoya, P. L., & Chivenge, P. (2017). Building climate change resilience through adaptation in smallholder farming systems in semi-arid Zimbabwe. *International Journal of Climate Change Strategies and Management*, 9(2), 151-165. <https://doi.org/10.1108/IJCCSM-07-2016-0092>

Research topic

Jiri et al. (2017) evaluated adaptation measures and their drivers among smallholder farmers in an arid district in southeastern Zimbabwe.

Methodology

The authors conducted a survey of 100 farmers in four wards, as well as focus group discussions and key informant interviews. They then compared adapters to non-adapters using descriptive statistics and used a regression model to assess predictors of adaptation, treating adaptation as a dichotomous variable (with adaptation defined as adoption of one or more specific practices, such as dry planting, use of short season varieties, and conservation agriculture). Predictor variables in their regression model did not include gender, but did include age of household head, extension access, number of work-capable individuals in household, access to credit, farm income, livestock holdings, land area, employment status, and literacy.

Relevant findings

In examining descriptive statistics of adapters and non-adapters, there was no significant difference in the rates that men and women reported adopting adaptations. However, younger, literate farmers with more farm income and access to credit appeared more likely to adapt than their counterparts. The authors’ regression model confirmed that access to credit, extension, and number of household members fit to work were positively associated with farmers’ decision to adapt to climate variability, while age of the household head was negatively associated. Gender is not discussed further, but the impacts of access to credit and extension may be relevant, although the authors do not discuss differential access to these resources in the study context.

Conclusions

The authors concluded that farmers in this setting had adapted by adjusting their farming practices to better respond to drought and climate risk, especially through on-farm diversification. However, they emphasized the need to support growth of non-farm livelihood options in rural areas.

Reviewer comment

Possibly relevant sections: access to finance, information/extension. Minimal attention to gender.

20. Khan, M. A., Hasan, K., & Kabir, K. H. (2022a). Determinants of households' livelihood vulnerability due to climate induced disaster in southwest coastal region of Bangladesh. *Progress in Disaster Science*, 15(May), 100243. <https://doi.org/10.1016/j.pdisas.2022.100243>

Research topic

Khan et al. (2022a) examined the factors influencing households' livelihood vulnerability to climate-related risks in three unions of coastal Bangladesh.

Methodology

The authors surveyed 300 household heads (>90% men) and used the Livelihood Vulnerability Index (incorporating sociodemographics, livelihood data, social networks, health, food, water, and exposure to climate variability and disasters) to assess household vulnerability. They also used a logit regression model to assess predictors of vulnerability.

Relevant findings

The authors found significant variation in vulnerability by location and identified numerous other predictors of vulnerability. They detected that female-headed households, those with more dependents, those whose head had not attended school, and those that did not grow crops were significantly more vulnerable in at least two out of three unions. Lower vulnerability scores were, at least in one location, associated with diversified livelihoods, having a family member working in a different community, access to community and government support, and membership in community groups. The authors attributed the heightened vulnerability among female-headed households to their "limited ability to adapt to extreme climate-related hazards and lower access to livelihood resources and strategies" (p. 11), but they do not engage further with gender norms.

Conclusions

The authors advocated for greater partnership between government entities and communities to improve the effectiveness of adaptation measures and their contributions to social inclusion. They recommended education programs, salinity-resistant crop breeding, and support for social networks.

Reviewer comment

Relevant sections: Resource access, income generation options. Very limited discussion of norms.

21. Khan, M. T. I., Anwar, S., Sarkodie, S. A., Yaseen, M. R., Nadeem, A. M., & Ali, Q. (2022b). Comprehensive disaster resilience index: Pathway towards risk-informed sustainable development. *Journal of Cleaner Production*, 366(June), 132937. <https://doi.org/10.1016/j.jclepro.2022.132937>

Research topic

Khan et al. (2022b) compared disaster resilience across country income levels.

Methodology

Khan et al. (2022b) develop a 9-dimension disaster resilience index incorporating measures of economic stability, infrastructure, digitalization, governance, agricultural development, women's empowerment, human capital, emergency capabilities, and human capital. They justified inclusion of women's empowerment by describing women's differential vulnerability, linked to "limited access to economic activities, information, social networks, skills, transportation, and decision-making" (p. 6), as well as women's unique roles in forging social connections and caring for the household post-disaster. Indicators for women's empowerment included access to justice, women's participation in CSOs, % of employers who were female, freedom of women from forced labor, freedom of discussion, freedom of domestic movement, % female labor force participation, % female self-employment, and % representation in parliaments.

Relevant findings

The authors calculated index scores for 91 countries and offered comparisons across country income levels. The authors' inclusion of women's empowerment indicators broadly indicate recognition of the role of norms in shaping societal resilience to disasters, but they offer no in-depth discussion of this. They found that resilience scores paralleled income levels, with highest-income countries the most resilient.

Conclusions

They concluded by recommending resilience-boosting measures, including health infrastructure, support for agricultural and financial sectors, enhanced human and social capital, institutional reforms, and "women's participation in society, access to employment, security, and the creation of self-employment opportunities for women" (p. 12).

Reviewer comment

Relevant sections: behavioral norms, livelihood generation options.

22. Khatri-Chhetri, A., Regmi, P. P., Chanana, N., & Aggarwal, P. K. (2020). Potential of climate-smart agriculture in reducing women farmers' drudgery in high climatic risk areas. *Climatic Change*, 158, 29-42. <https://doi.org/10.1007/s10584-018-2350-8>

Research topic

Khatri-Chhetri et al. (2020) sought to identify climate-smart agriculture (CSA) practices with potential to reduce women's labor burdens in Nepal.

Methodology

They first used geospatial analysis to identify hotspots for poverty, climate risk, and women's engagement in agriculture, then selected two districts for an in-depth assessment of women's roles and responsibilities. For this, they conducted a survey of 215 households and compared men's and women's labor contributions to different agricultural activities. They also conducted a literature review around gender and CSA, organized focus group discussions with women farmers' groups, and created an assessment index to identify CSA practices and technologies with potential to reduce women's drudgery, either through increased productivity or income.

Relevant findings

The authors found women's contributions (in number of days) to be significantly higher than men's, in part due to male out-migration. They also found clear differentiation of responsibilities, with women handling rice transplantation, weeding, irrigation, and wheat harvest and threshing while men handled more land preparation, crop sowing, and fertilizer application. The authors used regression to identify predictors of women making greater labor contributions; women were more likely to be heavily involved in agriculture in households with limited landholdings, less off-farm income, larger family size, agriculture as a primary occupation, and female household headship. The authors' efforts to identify gender-sensitive CSA practices yielded several recommendations, including direct rice seeding, use of zero tillage machines, laser land leveling, and green manure.

Conclusions

The authors concluded that numerous CSA practices could reduce women's labor burdens, especially if targeted to identified hotspots of vulnerability. They also stressed CSA's ability to increase women's access to resources, involvement in farming decisions, and market opportunities.

Reviewer comment

Relevant sections: division of labor, access to land, access to diverse livelihoods, male outmigration. [This doesn't address social norms head-on, but clearly touches on related issues.]

23. Kjeldsberg, C., Shrestha, N., Patel, M., Davis, D., Mundy, G., & Cunningham, K. (2018). Nutrition-sensitive agricultural interventions and gender dynamics: A qualitative study in Nepal. *Maternal and Child Nutrition*, 14(3), 1-9. <https://doi.org/10.1111/mcn.12593>

Research topic

Kjeldsberg et al. (2018) assessed the gendered impacts of a nutrition-focused agricultural development project in Nepal.

Methodology

The authors followed up on a nutrition project that had targeted women through a training-of-trainers approach and provided basic inputs to support homestead gardening and poultry production, alongside nutrition and hygiene trainings. Four years after completion of the project, the author conducted ten focus group discussions with women in two target districts to qualitatively evaluate project impacts on women's decision-making power, freedom of mobility, social support, workload and time, and self-efficacy.

Relevant findings

The authors found that women who had participated in project trainings and received livelihoods support later reported increased self-efficacy, respect from household members and inclusion in financial decision-making, and confidence selling in markets, which contributed to income generation. Limits to women's engagement in decision-making and financial control emerged as an issue in one district, linked to women's limited power as knowledge-holders and conservative social norms that emphasize women's supportive household roles relative to men's leadership roles. In one district, remoteness undermined access to markets and efficacy of trainings. The women cited workload and time as key constraints to continued engagement in, and income generation through, gardening and poultry, given their existing household responsibilities.

Conclusions

The authors concluded that women had derived numerous benefits from the nutrition project, but that barriers and facilitating factors differed across locations due to varied sociocultural dynamics, especially within households. They advocated for gender-sensitive nutrition interventions that address gender-related barriers and household dynamics directly.

Reviewer comment

Relevant sections: decision-making / norms around subservience, time/labor constraints; also touches on financial control.

24. Koudjom, E. (2022). Climate change adaptation and maize productivity: a gender-based analysis. *Climatic Change*, 173(1-2). <https://doi.org/10.1007/s10584-022-03403-1>

Research topic

Koudjom (2022) assessed productivity levels in maize systems and the relationships between productivity and climate change adaptation.

Methodology

The authors first used a generalized selection model to identify factors associated with adaptation at the farm level, treated as a dichotomous variable. They then used plot-level data on 8622 maize plots in 6171 households from the Harmonized Household Living Conditions Survey in Togo, comparing productivity between men's, married women's, and unmarried women's plots.

Relevant findings

The author found that the core drivers of adaptation were "secondary education, maize farmer assets, and climate information for both genders" (p. 8). More men reported implementing adaptive measures (defined as one or more of a set of practices, such as improved seed use and off-season cropping). The author also found numerous differences between men and women in the sample, with men demonstrating greater use of agricultural inputs (labor, fertilizer, and improved seed), more education, and better access to extension and climate information. While across genders, farmers who adapted to climate change had more productive plots than those who did not, the productivity of men's plots appeared to increase more than women's productivity when adaptation measures were in place. Among 'adapters,' men's plots were 23.5% more productive than married women's, whose plots were in turn 28.5% more productive than unmarried women's. The authors attributed these differences to structural

advantages for men and disadvantages for women (linked to use of improved seeds, off-season cropping, and the combination of strategies, as well as secondary education, producer assets).

Conclusions

The authors concluded that there is need to ensure more equal access to education, land, productive inputs, and climate information. They emphasized the need for improved training, awareness-raising, and extension related to adaptive practices, alongside strengthened input supply and financial support systems.

Reviewer comment

Relevant sections: access to inputs/resources, education, intersectionality.

25. Kpienbaareh, D., Kansanga, M. M., Yiridoe, E., & Luginaah, I. (2022). Exploring the drivers of herbicide use and risk perception among smallholder farmers in Ghana. *Gender, Technology and Development*, 0(0), 1-27. <https://doi.org/10.1080/09718524.2022.2092939>

Research topic

Kpienbaareh et al. (2022) examined backyard farmers' perceptions of and motivations in using herbicides in Ghana.

Methodology

The authors used participatory photovoice (i.e., allowing participants to document their lives with photos for three weeks) and follow-up interviews with 48 backyard farmers (10 female) who used herbicides, sampled in two districts in semi-arid northern Ghana. Participants selected their preferred photos, which were printed for discussion facilitation in two cross-community focus groups. The authors also conducted several key informant interviews with local agrodealers and government representatives.

Relevant findings

Kpienbaareh et al. (2022) highlighted that although herbicide spraying is among the tasks considered "men's work," women in this context were unique vulnerable to the health impacts of herbicides, as they often handle manual labor in the field after herbicide application. All women interviewed reported using herbicides first and foremost to avoid costs related to tractors and labor (although this was also the most common response among men), and cited climate variability as a tertiary concern (although at higher rates than men). The authors also found differentiated perceptions of herbicides' impacts. More women than men took pictures of community dams and bodies of water to which herbicide runoff flowed—a concern likely linked to women's role in collecting water for the household—or of poor seed germination—likely linked to women's role in sowing and reseeding fields.

Conclusions

The authors concluded that herbicide use is motivated by intersecting economic, environmental, and socio-cultural factors, and that perceptions of herbicides' impacts vary by gender. They advocated for gender mainstreaming in agricultural development to address underlying disparities in access to resources.

Reviewer comment

Relevant sections: divisions of labor.

26. Krell, N. T., Giroux, S. A., Guido, Z., Hannah, C., Lopus, S. E., Caylor, K. K., & Evans, T. P. (2021). Smallholder farmers' use of mobile phone services in central Kenya. *Climate and Development*, 13(3), 215-227. <https://doi.org/10.1080/17565529.2020.1748847>

Research topic

Krell et al. (2021) examined smallholders' access to mobile phones and use of mobile services in agriculture.

Methodology

The authors drew on a survey of 577 farming households across thirty-five communities in central Kenya that captured sociodemographics, farmer group membership, and phone access and use. They then used descriptive statistics and regression models to understand patterns in mobile phone use.

Relevant findings

Krell et al. (2021) found almost universal mobile phone ownership (although only 35% of farmers owned smartphones, and more women than men owned only 'basic' phones). However, only a quarter of phone owners used it to access information about agriculture, livestock, or markets. Mobile services' use was found to be associated with men, farmers with more formal education (a particularly strong correlation), informal farmer group membership, and ownership of household and livestock assets. Women cited a more comprehensive set of barriers to mobile service use than men and were less aware and knowledgeable than men about how to use these services. The authors also argued that women's financial dependence on men and limited technical literacy constrained their access to and use of phones.

Conclusions

The authors concluded that mobile phone-based services in agriculture still do not seem to be designed for rural, low-income, and less educated individuals, including women. These services are increasingly designed for more advanced phones which remain inaccessible to many. The factors the authors identified as significant in predicting use of mobile services (including education, farmer group membership, and smartphone ownership) interact with gender to limit women's access to these services.

Reviewer comment

Relevant sections: access to information, access to finance, education, poverty.

27. Leal Filho, W., Totin, E., Franke, J. A., Andrew, S. M., Abubakar, I. R., Azadi, H., Nunn, P. D., Ouweneel, B., Williams, P. A., & Simpson, N. P. (2022). Understanding responses to climate-related water scarcity in Africa. *Science of the Total Environment*, 806, 150420. <https://doi.org/10.1016/j.scitotenv.2021.150420>

Research topic

Leal Filho et al. (2022) sought to understand the scope of water scarcity challenges in Africa as well as common responses.

Methodology

The authors first used geospatial modeling to identify areas facing water scarcity and the scale of human impacts, then reviewed secondary data (from GAMl) on adaptation and conducted a supplementary systematic literature review to understand adaptive responses to water stress.

Relevant findings

Leal Filho et al. (2022)'s discussion of gender was not related to their primary (geospatial) data but rather to their literature review. The authors described several aspects of social vulnerability as relevant to their analysis, including the role of gender: "Social capital, poverty, and gender depress women's agency and pathways to overcome water scarcity in rural areas (Pearson et al., 2015), particularly when household structures and social norms entrench patriarchal arrangements such as those of title and tenure (Rao et al., 2019b)" (p. 6). The authors also noted that men's emigration can burden women who lack access to social support, and that women often serve as primary water collectors and thus face the brunt of water scarcity's impacts. They found that most planned responses to water scarcity, which generally focused on coping strategies, food, shelter, and livelihoods, intentionally targeted low-income community, with the largest share of these giving deliberate attention to women.

Conclusions

The authors advocated for attention to the "social capital, institutional, livelihoods and wellbeing dimensions" of water scarcity rather than strictly technical supply and management (p. 6). They also stressed the need to integrate responses to water scarcity with those related to food security, urbanization, desertification, and security.

Reviewer comment

Relevant sections: access to resources, behavioral norms.

28. Maguire-Rajpaul, V. A., Khatun, K., & Hiron, M. A. (2020). Agricultural Information's Impact on the Adaptive Capacity of Ghana's Smallholder Cocoa Farmers. *Frontiers in Sustainable Food Systems*, 4(March). <https://doi.org/10.3389/fsufs.2020.00028>

Research topic

Maguire-Rajpaul et al. (2020) explored the impacts of four types of agricultural information on different types of farmers' adaptive capacity.

Methodology

The authors conducted 12 exploratory focus group discussions that informed 48 interviews with cocoa farmers (one-third of whom were women) and several key informant interviews in two regions of south-central Ghana. Interviews covered coping strategies farmers employed in response to a recent drought, challenges encountered, and knowledge-sharing practices. Recognizing that "adaptation is a social process—rather than solely a technical challenge" (p. 3), the authors sought to uncover the social differentiation of adaptive capacity.

Relevant findings

Maguire-Rajpaul et al. (2020) described how women in this context must balance domestic responsibilities with farm labor and are less likely to own land, which limits their engagement in trainings and information programs, farmers' organizations, and credit programs. While they found that information exposure generally increased farmers' adaptive capacity, women faced unique challenges to accessing information. They were less likely to engage with extension and received, gave, and observed fewer adaptive technologies from and with fellow farmers. Men's control over cropping decisions reportedly limited women's on-farm adaptation options, while social norms limited their off-farm adaptation options: "Choices for off-farm income diversification are shaped by local supply and demand, but female interviewees described how a more decisive factor is locally defined socio-cultural norms dictating which alternate sources of income are "appropriate" for women" (p. 12). Despite the small sample size, women reported taking significantly fewer adaptive actions, were significantly less likely to report awareness of planting shade trees as an adaptive action, and were significantly more likely to say their production was worse 18 months after the drought than before it. Respondents with little or no education also fared worse.

Conclusions

The authors concluded that information treatments had helped cocoa farmers in this context to adapt, but that most adaptive strategies employed were short-term coping measures rather than transformative, resilience-building changes. However, they found that socioeconomic and structural conditions mediated the impact of information received, preventing some farmers from adopting practices about which they learned.

Reviewer comment

Relevant sections: information, education, extension, income generation options, decision-making.

29. Malherbe, W., Sauer, W., & Aswani, S. (2020). Social capital reduces vulnerability in rural coastal communities of Solomon Islands. *Ocean and Coastal Management*, 191(March), 105186. <https://doi.org/10.1016/j.occoaman.2020.105186>

Research topic

Malherbe et al. (2020) examined the vulnerability of coastal communities in the Solomon Islands with specific attention to social capital.

Methodology

Malherbe et al. (2020) developed a vulnerability assessment using the five capitals as their primary metrics. They surveyed 110 individuals (25% female) in five communities, using the aggregated data to compare conditions across communities and generate an overall vulnerability score across the study area. In this analysis, they emphasized social capital: community cohesion, gender equity, community engagement in decision-making, leadership, and equal resource access.

Relevant findings

The authors reported that gender roles in this context were firmly rooted in tradition and the church; women occupied largely domestic spaces and engaged in some fishing and gardening, while men undertook primarily fishing-related tasks and heavy physical labor. Roughly half of participants felt that women's control over livelihoods and resources was at least somewhat unequal. While respondents often reported that women served as community leaders, these roles were primarily in the household, churches, or women's organizations. The authors reported that social capital contributed to community resilience and was bolstered most by community cohesion, gender equity, leadership, equal access to services and resources and decision making. The authors stated that access to resources is generally equal but low, then suggest it may not be equal for all groups (but gender is not mentioned). The authors also suggested that too much external influence, including the transition to a cash-based economy, could undermine traditional social structures and increase vulnerability.

Conclusions

The author advocated for policy measures to preserve social capital, especially by bolstering local cultural traditions. They cautioned against undue disruption of traditional norms that maintained social cohesion in the studied communities—including women's involvement in non-traditional livelihoods that might lead to discrimination and harassment.

Reviewer comment

Relevant sections: behavioral norms, income generation options. The arguments in this paper are difficult to understand; they present gender equality as beneficial to social capital and resilience, then discourage disruption of traditional gender norms.

30. Melvani, K., Bristow, M., Moles, J., Crase, B., & Kaestli, M. (2020). Multiple livelihood strategies and high floristic diversity increase the adaptive capacity and resilience of Sri Lankan farming enterprises. *Science of the Total Environment*, 739, 139120. <https://doi.org/10.1016/j.scitotenv.2020.139120>

Research topic

Melvani et al. (2020) studied traditional farming enterprises in Sri Lanka to understand which factors contributed to resilience and adaptive capacity.

Methodology

The authors used surveys and biophysical observations from 85 farm enterprises to assess land uses, farmer perspectives on climate variability, and socioeconomic characteristics. The authors then used ANOVAs to evaluate whether plant and animal diversity and other variables were associated with land use, location, and farmer characteristics.

Relevant findings

Melvani et al. (2020) found that forest gardens were a preferred land use type in this context, providing resilience to farming systems and continuous but diverse benefits to households. Female-managed Forest gardens had higher species richness and diversity than male-managed gardens, which the authors attributed to women's responsibility for maintaining household food security and nutrition. Age and education level were not significant predictors of biodiversity in forest gardens. Aside from the identified difference in biodiversity between male- and female managed plots, the authors did not discuss gender relations or their role in shaping adaptive capacity.

Conclusions

The authors concluded that forest gardens held the most adaptive capacity of all land use types in the study in terms of diversity and functional utility. They argued that traditional knowledge and skills were critical to building climate resilience.

Reviewer comment

Relevant sections: behavioral norms / division of labor. Minimal discussion of gender.

31. Mogomotsi, P. K., Sekelemani, A., & Mogomotsi, G. E. J. (2020). Climate change adaptation strategies of small-scale farmers in Ngamiland East, Botswana. *Climatic Change*, 159(3), 441-460. <https://doi.org/10.1007/s10584-019-02645-w>

Research topic

Mogomotsi et al. (2020) documented adaptation strategies among smallholder farmers in Botswana.

Methodology

The authors conducted a survey of 300 heads of farming households (186 female, 114 male, non-randomized) and key informant interviews. Surveys captured household demographics, farming behaviors, knowledge and perceptions of climate change, adaptation strategies, and challenges encountered. The authors used a probit regression model to assess predictors of adaptation (as a dichotomous variable) and SWOT analysis to evaluate local institutions' role in adaptive processes.

Relevant findings

Mogomotsi et al. (2020) found that 84% of farmers were implementing at least one adaptive strategy, most of which were rooted in indigenous knowledge. The most popular adaptations were cultivation of fast-maturing crops and provision of supplementary feed for livestock. The barriers to adaptation that respondents most frequently cited were access to capital, inputs, and water; indeed, in the authors' regression model, they found a strong positive correlation between household income and adaptation. Older, more experienced, and better educated farmers in smaller households and those with more awareness of climate concerns were also more likely to have adaptive strategies in place. Gender of the household head was not a significant predictor of adaptive behaviors. However, key informants highlighted that extension services were most accessible to better-resourced farmers. The authors did not engage directly with gender norms but many of the predictors of adaptation in their model have potential relevance.

Conclusions

The authors concluded that there are many sociodemographic factors associated with farmers' adaptation. Given the importance of information and awareness, they advocated for government sensitization campaigns and engagement between institutions and local communities with important existing knowledge.

Reviewer comment

Relevant sections: possibly access to finance, inputs, education, extension. No direct discussion of norms and very little on gender.

32. Nixon, R., Ma, Z., Zanotti, L., Khan, B., Birkenholtz, T., Lee, L., & Mian, I. (2022). Adaptation to Social-Ecological Change in Northwestern Pakistan: Household Strategies and Decision-making Processes. *Environmental Management*, 69(5), 887-905. <https://doi.org/10.1007/s00267-021-01583-7>

Research topic

Nixon et al. (2022) assessed household adaptive strategies in Pakistan, including the values, trade-offs, and structural factors influencing choice of adaptations. They acknowledged that "existing power dynamics within and among households and communities can shape whose values are prioritized in adaptation decision-making" (pp. 888-889).

Methodology

To explore this topic, the authors conducted and analyzed twenty-five semi-structured interviews with household heads and community leaders and paired these with secondary data from reports related to socioecological change in the region. Only two interviews were held with women because many male household heads refused permission for their wives to participate, and

other women were unwilling to speak with researchers. Interview data around adaptation were coded to identify common conceptual trends.

Relevant findings

The authors found that adaptations practiced in this context related to environmental management, livelihood diversification, migration, community pooling, and changes to marketing practices. Values that influenced households' adaptation decisions included those related to tradition, security, and economics (e.g., labor, time, and finance). Trade-offs between these values were often required (e.g., investing financial resources to support environmental management, or abandoning traditional livelihoods for more profitable ones). These processes were also influenced by multi-scalar factors beyond the households (e.g., regional environmental projects or availability of alternative livelihoods). However, the authors could draw no specific insights on the gender dynamics of adaptation or adaptive capacity.

Conclusions

The authors concluded that adaptations are location-specific and influenced by values and trade-offs between them (not strictly economics), as well as multi-scalar factors outside the household. They emphasized the need to appeal to core values, and understand whose values are prioritized and how values might shift over time, in designing adaptation interventions.

Reviewer comment

Relevant sections: Decision-making. No specific insights on gender, as the authors could not actually talk with women.

33. Nkuba, M., Chanda, R., Mmopelwa, G., Kato, E., Mangheni, M. N., & Lesolle, D. (2019). The effect of climate information in pastoralists' adaptation to climate change: A case study of Rwenzori region, Western Uganda. *International Journal of Climate Change Strategies and Management*, 11(4), 442-464. <https://doi.org/10.1108/IJCCSM-10-2018-0073>

Research topic

Nkuba et al. (2019) sought to understand the impact of indigenous forecasts (IFs) and scientific forecasts (SFs) on pastoralists' adaptive behaviors.

Methodology

The authors conducted two focus group discussions and household surveys with 270 pastoralists (59% male) in western Uganda. They focused the surveys on household characteristics, use of IFs and SFs, adaptation measures, and resource access. Using a multivariable probit model, they assessed relationships between predictor variables and use of different adaptive strategies.

Relevant findings

Nkuba et al. (2019) found households using only IFs were more often engaged in non-farm enterprises and livestock sales, and typically had greater assets and more education. Use of both IFs and SFs was associated with livestock migration (therefore, mobile pastoralism) and more limited assets. Men were significantly more likely to practice herd mobility (i.e., short-term grazing over long distances—a "very strenuous activity" (p. 456)), but no further gender-based differences were evident. Access to extension and education were associated with adaptation broadly, while access to inherited land and hired labor enabled specific adaptive measures—e.g., land access increased the likelihood of engaging in non-farm enterprises, well excavation, pasture development, and improved livestock management, leading to increased market participation and income generation. The authors did not engage further with gender, although their findings may have relevant implications.

Conclusions

The authors concluded that use of IFs only was associated with a focus on non-weather-related activities and more sedentary pastoralism, while use of both SFs and IFs was associated with mobile pastoralism. They did not offer much further interpretation of their findings but advocated for increased investment in weather services.

Reviewer comment

Relevant sections: access to land, extension, education, information.

34. Ortiz, R., & Peris, J. (2022). The Role of Farmers' Umbrella Organizations in Building Transformative Capacity around Grassroots Innovations in Rural Agri-Food Systems in Guatemala. *Sustainability (Switzerland)*, 14(5). <https://doi.org/10.3390/su14052695>

Research topic

Ortiz & Peris (2022) examined how farmers' organizations support families' transformative capacity to build resilient agri-food systems in western Guatemala.

Methodology

The authors conducted interviews with fourteen individuals and focus group discussions (including with women farmers, where the focus was on women's engagement in local group activities, decision-making, and capacity development). They linked transformative capacity to agency and interaction forms (inclusive governance and leadership), development processes (sustainability and system awareness), knowledge and reflexive learning, and relational dimensions from individual and household scale to regional and national scale.

Relevant findings

Ortiz & Peris (2022) found that umbrella farmer organizations played a crucial role in supporting the scale-out of transformative innovations, in allowing for collective visioning and development of, experimentation with, and promotion of new technologies. However, they found elite capture of leadership in these institutions to be a common problem affecting women; women encountered resistance when taking leadership positions and had trouble "speaking loudly" in discussions with men (p. 14). Women-led agriculture and women-focused projects were not perceived to be government priorities, but some local farmers groups (specifically climate-adapted villages) had prioritized women's leadership, were creating new opportunities for women, and built their capacity and confidence. These groups also created opportunities for more farmer engagement in decision-making, communities of practice, and experimentation, broadly.

Conclusions

The authors concluded that while umbrella farmers' organizations have huge potential to support the scale-out of transformative technologies and practices, there is need for more attention to gender and inclusion to maximize their transformative potential.

Reviewer comment

Relevant sections: social capital, behavioral norms, decision-making

35. Ouédraogo, M., Barry, S., Zougmore, R. B., Partey, S. T., Somé, L., & Baki, G. (2018). Farmers' willingness to pay for climate information services: Evidence from Cowpea and Sesame producers in Northern Burkina Faso. *Sustainability (Switzerland)*, 10(3). <https://doi.org/10.3390/su10030611>

Research topic

Ouédraogo et al. (2018) assessed sesame and cowpea farmers' willingness to pay (WTP) for different forms of climate information services in Burkina Faso.

Methodology

The authors surveyed 170 farmers (113 men, 56 women) in the Yatenga block in the country's North, collecting household and farm characteristics as well as WTP using open-ended contingent valuation questions. Their analysis included Tobit models to assess predictors of WTP for different services.

Relevant findings

Ouédraogo et al. (2018) found some differences in farmers' WTP for specific types of climate information, with seasonal forecasts and daily weather information most in demand. They also found that men, younger farmers, less educated farmers, and those with higher awareness of climate information generally had higher WTP, while the size of the household had some impact on the type of climate information preferred. The authors do not offer any explanation of gendered patterns in WTP, nor do they discuss gender at any length.

Conclusions

The authors conclude that WTP is influenced by the type of climate information and farmer characteristics.

Reviewer comment

Relevant sections: information. No substantial discussion of gender norms.

36. Paudel, D., Tiwari, K. R., Raut, N., Bajracharya, R. M., Bhattarai, S., Sitaula, B. K., & Thapa, S. (2022). What affects farmers in choosing better agroforestry practice as a strategy of climate change adaptation? An experience from the mid-hills of Nepal. *Heliyon*, 8(6), e09695. <https://doi.org/10.1016/j.heliyon.2022.e09695>

Research topic

Paudel et al. (2022) examined uptake of traditional and improved agroforestry practices in three districts of Nepal.

Methodology

The authors conducted 420 household surveys, 24 key informant interviews, and focus group discussions, and then used regression to evaluate predictors of adaptation via agroforestry. Surveys captured sociodemographics, perceptions of climate events and responses to them, use of agroforestry practices, and primary motivations behind the use of agroforestry.

Relevant findings

Paudel et al. (2022) documented a range of adaptations, although agroforestry was not among the most popular—more common were on-farm adaptations and crop diversification tactics, as well as income diversification and emigration. They found older, more educated farmers and those growing cash crops were more likely to adapt in general. Older farmers, females, farmers generating income from tree products, those from larger households, those with smaller landholdings, and those better informed about climate change were more likely to adopt improved agroforestry practices. A greater proportion of females than males had participated in agroforestry trainings, but men reported access to a wider set of information sources (e.g., community meetings). Women's relatively deep engagement with agroforestry is not uncommon, the authors argued, and likely accelerated by their growing responsibilities on-farm in the context of men's increasing emigration.

Conclusions

The authors argue that limited awareness of climate change in this setting constrained adaptation, including through agroforestry. They advocate for expanded education around agroforestry, especially as it pertains to climate change.

Reviewer comment

Relevant sections: access to information, migration.

37. Porcher, V., Carrière, S. M., Gallois, S., Randriambanona, H., Rafidison, V. M., & Reyes-García, V. (2022). Growing up in the Betsileo landscape: Children's wild edible plants knowledge in Madagascar. *PLoS ONE*, 17(2 February), 1-20. <https://doi.org/10.1371/journal.pone.0264147>

Research topic

Porcher et al. (2022) assessed variation in local ecological knowledge of wild edible plants in Madagascar, specifically examining children's and gendered knowledge.

Methodology

The authors collected individuals' ethnobotanical knowledge through free-listing interviews with adults (n = 42, including 22 women) and children (n = 42, including 22 girls) among the Betsileo population in biodiverse south-central Madagascar. They also held semi-structured interviews and forest walks with five key informant adults and three children to discuss species names,

consumption habits, and locations. They then calculated cognitive salient indices for interviewed individuals to assess variation in knowledge among groups by gender and age and examined identified species' characteristics.

Relevant findings

Porcher et al. (2022) reported on the high gendered nature of social organization in Betsileo communities. While men cared for animals, prepare rice fields, and handle activities that require mobility (e.g., gathering, hunting, fishing, and trading), women's responsibilities were focused on the homestead (e.g., managing the rice nursery, home garden, and other crops, as well as childcare). The wild edible plants that women and men could identify were significantly different (although similar in total number), as were those that children and adults could identify. This suggested differentiated knowledge based on varied day-to-day experiences and relationships with nature; women and girls were more likely to identify plants growing in habitats near the homestead that could be eaten as a whole and herbaceous plants, all of which were key components of foods prepared by women. Men knew more endemic plants growing in remote pastures and undisturbed habitats.

Conclusions

The authors concluded that knowledge of wild edible plants is differentiated both by gender and life stage. Gender appeared to influence the specialization of this knowledge over a lifetime of gender-differentiated daily experiences.

Reviewer comment

Relevant sections: intersectionality, division of labor/behavioral norms.

38. Quiros, T. E. A. L., Beck, M. W., Araw, A., Croll, D. A., & Tershy, B. (2018). Small-scale seagrass fisheries can reduce social vulnerability: a comparative case study. *Ocean and Coastal Management*, 157(February), 56-67. <https://doi.org/10.1016/j.ocecoaman.2018.02.003>

Research topic

Quiros et al. (2018) sought to understand the dynamics of social vulnerability of seagrass fisheries in the Philippines.

Methodology

The authors compared the sensitivity and adaptive capacity of two small-scale seagrass fisheries. They leveraged data from fisher (n = 160) and household surveys (n = 162), underwater surveys, landing surveys, and participant observation, in order to assess natural capital and socioeconomic conditions (demographics, standard of living, reliance on fishery-based livelihoods, engagement of different household members in seagrass-related activities, and resource use patterns).

Relevant findings

The authors found that in seagrass fisheries in the Philippines, women played an important role; although they comprised a minority of fishers (13% and 15% in the two study regions), they represented the majority of daytime gleaners in seagrass beds (62% and 71%). Women gleaners contributed 15% and 21% of the total catch in each study location, which also improved community-level adaptive capacity due to the diversified income source they provided. Women also engaged in processing activities and local sales, which contributed to household income. However, the authors determined that women's limited role as gleaners made them uniquely vulnerable to changes in seagrass resources; men, in contrast, had more diverse fishing gear, including boats, and were thus better able to diversify their fishing methods in response to environmental stressors. Adaptive capacity in this study was strongly tied to the availability of alternative incomes, although this was not discussed directly in relation to gender or social norms.

Conclusions

The authors concluded that sociodemographics (including gender and social capital) and economic factors (especially access to livelihoods, both fishing-related and beyond) were key factors influencing vulnerability in this system. They advocated for protection of seagrass habitats to slow observed ecological decline

Reviewer comment

Relevant sections: income generation options.

39. Rozas, L., & Busse, P. (2022). The foodscapes of children and adolescents attending schools in Lima, Peru. *Appetite*, 168(October 2020), 105738. <https://doi.org/10.1016/j.appet.2021.105738>

Research topic

Rozas & Busse (2022) evaluated how children and adolescents' eating behaviors in Peru were shaped by the interactions of physical, social, and institutional factors.

Methodology

The authors observed eating behaviors in one public and one private school in Lima over the course of one week. They paired these observational data with 44 interviews with parents, covering household food production and eating habits, school food services and diet, food marketing, and knowledge of healthy eating. They then used qualitative content analysis to assess patterns.

Relevant findings

Rozas & Busse (2022) reported that women in this context were primary caregivers and food preparers within households, and thus the individuals who established rules around children's eating, while men "spoiled" children with junk food on weekends. At school, girls received less lunch money and purchased food independently less often than boys, who reportedly also experienced more freedom and mobility to purchase food outside school. Girls' eating habits were more closely monitored than boys' by both parents and through social pressure during school mealtimes, although explanations for these gendered outcomes are not discussed at depth. Children in higher socioeconomic strata were exposed to a wider food environment outside the home (e.g., to grocery stores and restaurants).

Conclusions

Rozas & Busse (2022) concluded that children and adolescents' eating habits were influenced by a number of interacting factors, notably demographic characteristics such as economic status, age, and gender, which intersected with their individual agency, physical environment, and influence from peers and parents. The authors identified social relations (especially peer and parent influence) as little-studied determinants of eating habits in urban school-aged children.

Reviewer comment

Relevant sections: intersectionality, behavioral norms, mobility, decision-making, access to resources/finance.

40. Pathak, S., Panta, H. K., Bhandari, T., & Paudel, K. P. (2020). Flood vulnerability and its influencing factors. *Natural Hazards*, 104(3), 2175-2196. <https://doi.org/10.1007/s11069-020-04267-3>

Research topic

Pathak et al. (2020) examined household-level vulnerability to riverine floods in Nepal.

Methodology

The authors collected data from 217 households in one district via surveys that assessed sociodemographic characteristics, livelihood strategies, welfare outcomes, and flood experiences and responses. They then used an integrated vulnerability approach (combining biophysical and social factors of exposure, sensitivity, and adaptive capacity) to develop a household vulnerability index with 30 indicators derived from literature review and expert consultation. After calculating household vulnerability scores, the authors used an ordinal response model to identify factors associated with household vulnerability.

Relevant findings

Pathak et al. (2020) found that heightened household vulnerability was associated with 'conventional' adaptation measures (typically low cost and traditional knowledge-based), location in a river basin, and a 'lack of scientific measures to flood control.' Household factors associated with higher vulnerability included access to credit, migration, a higher household ratio of women to men, female household headship, having fewer livestock, and lower total income. The authors mentioned that women's greater presence in the household could contribute to vulnerability due to their limited mobility. Otherwise, gender norms were not discussed and little explanation of gender-related trends was offered, aside from a brief mention of women's limited decision-making authority.

Conclusions

The authors concluded that most households experienced moderate to high vulnerability and had low adaptive capacity, although they noted substantial variation across households and municipalities, attributed to socioeconomic differences and “relatively high exposure” (meaning unclear).

Reviewer comment

Relevant sections: Poverty. No direct discussion of gender norms.

41. Perez, L., Rios, D. A., Giraldo, D. C., Twyman, J., Blundo-Canto, G., Prager, S. D., & Ramirez-Villegas, J. (2020). Determinants of vulnerability of bean growing households to climate variability in Colombia. *Climate and Development*, 12(8), 730-742.
<https://doi.org/10.1080/17565529.2019.1685931>

Research topic

Perez et al. (2020) identified determinants of climate vulnerability among farming households in Colombia.

Methodology

Pairing data from 567 bean-growing households in four municipalities of Colombia with geospatial climate data, the authors developed a household-level vulnerability index using the IPCC framework (exposure, sensitivity, and adaptive capacity). Household surveys captured sociodemographic and farm information, perceptions of climate risk, and gender-related variables (number of females working on the farm, making productive decision, and engaged in pre-sowing, sowing, control, harvest, and post-harvest activities). Climate data captured the number of consecutive dry days. After scoring and categorizing the households into three groups (low, medium, and high vulnerability), the authors used a generalized ordered probit model to identify the factors determining households' placement in vulnerability categories.

Relevant findings

Perez et al. (2020) found that geography was a key predictor of vulnerability classification. Higher vulnerability was also associated with access to climate information (presumably a non-causal correlation), indebtedness, female headship, and, curiously, a higher percentage of both male and female household members being involved in decision-making (attributed to difficulty reaching consensus). Lower vulnerability was correlated with access to agronomic training, on-farm diversification, married household heads, more education, and at least one household member working outside the farm. The authors do not engage directly with issues around gender norms or offer much explanation of gender-related trends.

Conclusions

The authors concluded that vulnerability was highly variable and influenced by household characteristics and climate factors, and that the most vulnerable areas should be targeted with interventions.

Reviewer comment

Relevant sections: Possibly information/training, education, livelihood diversification. No direct discussion of gender norms.

42. Rao, N., Singh, C., Solomon, D., Camfield, L., Sidiki, R., Angula, M., Poonacha, P., Sidibé, A., & Lawson, E. T. (2020). Managing risk, changing aspirations and household dynamics: Implications for wellbeing and adaptation in semi-arid Africa and India. *World Development*, 125, 104667.
<https://doi.org/10.1016/j.worlddev.2019.104667>

Research topic

Rao et al. (2020) evaluated adaptive responses to environmental stressors in purposively selected semi-arid contexts experiencing climate variability.

Methodology

The authors undertook a large-scale comparative study across Ethiopia, Ghana, India (Karnataka and Tamil Nadu), northern Kenya, Mali, and north-central Namibia. They conducted focus groups and several dozen life history interviews in each location to grasp the normative context and understand individuals' coping and adaptive strategies. They paired this qualitative data with survey data from over 3,000 households across study sites. In the process, the authors explicitly aimed at "moving beyond the counting of numbers of vulnerable men and women to unpacking relations of power, of inclusion and exclusion in decision-making" (p. 5).

Relevant findings

The authors argued that men's and women's "life-cycle position" within household structures and across locations mediates their vulnerability. They pointed out that female-headed households are typically perceived as being more vulnerable due to social norms limiting women's ownership of productive resources, decision-making authority, and access to support services, as well as their responsibility for both productive and reproductive tasks. However, women's intersectional identities within this group influences their vulnerability.

The authors found that key adaptive responses in study regions were livelihood diversification, migration, changing agricultural practices, and leveraging support mechanisms, each detailed below. 1) Livelihood diversification was highly gendered, mediated by resource access as well as social norms. While older men had ample autonomy over livelihood decisions, young women and men were more constrained and engaged more often in migration, where urban livelihood options were highly gendered (with women limited to domestic work and petty trade, primarily). In India, although both young men and women aspired to off-farm work, young men had more opportunities to pursue it. Women's need to diversify livelihoods was also found to have potential to push them into risky work (e.g., sex work) when other options are unavailable. Younger women's ability to generate income sometimes led to greater agency and involvement in household financial decisions, but for older married women, diversification simply increased labor burdens. 2) Men were generally more likely to migrate than women due to social norms and the availability of "appropriate work," and younger women were more likely to migrate than older women. Men's migration often placed burdens of labor and responsibility on women left behind, who lacked the skills for wider work and for whom working outside the village was considered socially inappropriate. It did not necessarily lead to women's greater control over household resources. The impacts of remittances varied by household structure—remittances did not reach all female household members equally in polygamous households. 3) Changes to on-farm practices varied by gender, marital status, residential status, age, and education. Married women with access to land through their spouse appeared to have more options, while women borrowing land either changed varieties, used mixed cropping, or started composting. Younger women (and men) focused more on off-farm livelihoods. Increases in cash cropping and crop diversity increased agricultural labor for women but also led men to engage more in domestic work, in some places. 4) Social support mechanisms included community-based systems, but decision-making in these structures is dominated by older and wealthier participants. Marginalized young men were leaving home and going astray, making marriage less secure for young women, who turned to women's and self-help groups and sometimes built "matrifocal" families.

The authors also examined the dynamics of wellbeing outcomes. Both female-headed households and married women were generally found to be less food secure—the former due to resource constraints but the latter due to social pressures to keep the rest of the family food secure. Young women were especially vulnerable to food insecurity and poor wellbeing outcomes due to their heavy workloads and very limited decision-making authority. Their aspirations were, in many settings, essentially moot.

Conclusions

The authors concluded that gender relations and intersectional identities shaped adaptive responses in many ways across study sites. "Across our study sites, we find that gender divisions of labor are shifting, with women taking on more responsibility for managing household production, traditionally male activities, and household reproduction. Yet, gendered norms around what is socially appropriate or not shape the choices available to women and men within households, and these in turn are shaped by their specific social positions within their households and communities" (pp. 11-12)... "Social norms do restrict the choices available to women, yet the opportunities available to them are shaped both by the context and the quality of relationships in which they are embedded" (p. 12).

Reviewer comment

Relevant sections: (most of them): intersectionality, income generation options, migration, access to resources (land), domestic responsibilities/household roles, decision-making.

43. Sadiq, M. A., Kuwornu, J. K. M., Al-Hassan, R. M., & Alhassan, S. I. (2019). Assessing maize farmers' adaptation strategies to climate change and variability in Ghana. *Agriculture (Switzerland)*, 9(5), 1-17. <https://doi.org/10.3390/agriculture9050090>

Research topic

Sadiq et al. (2019) evaluated adaptive strategies and predictors thereof for maize farmers in eastern Ghana.

Methodology

The authors conducted surveys with 150 households (27% female-headed), then used regression to identify factors associated with specific adaptations (which they categorized as no adaptations, recommended practices, adoption of improved varieties, and soil-related strategies). They also conducted several focus group discussions.

Relevant findings

Sadiq et al. (2019) found that adaptation via recommended practices (including land rotation, spacing, tree planting) was associated with perceptions of rainfall, access to credit, and farming experience. Adaptation via soil management (including use of fertilizer, mounds/ridges, insecticides) was associated with male-headed households and rainfall perceptions, while adoption of improved crop varieties (including livestock rearing and mixed cropping) was associated with perceptions of rainfall and farming experience. Focus group discussions identified key constraints to adaptation, namely: access to credit, access to drought-tolerant varieties, low literacy, high costs of labor, access to inputs, and access to information. The authors did not discuss the role of gender in the study context.

Conclusions

The authors concluded that farmers practiced numerous adaptive strategies, which were each associated with specific household and farm characteristics. They advocated for wider provision of improved seed and irrigation infrastructure.

Reviewer comment

Relevant sections: Minimal engagement with gender. Possibly: access to credit/finance, access to inputs, information, education.

44. Sam, A. S., Padmaja, S. S., Kächele, H., Kumar, R., & Müller, K. (2020). Climate change, drought and rural communities: Understanding people's perceptions and adaptations in rural eastern India. *International Journal of Disaster Risk Reduction*, 44(December 2019). <https://doi.org/10.1016/j.ijdrr.2019.101436>

Research topic

Sam et al. (2020) explored perceptions of drought and adaptive strategies in a state in eastern India.

Methodology

The authors surveyed 689 households (90% headed by men) across five villages about sociodemographics, farm characteristics, and experiences with drought, and paired these with focus group discussions focused on coping and adaptive strategies. They used a multinomial logit model to assess predictors of the top five adaptive behaviors, with reduced food consumption as the base category.

Relevant findings

Sam et al. (2020) found that, generally, larger households, those with older household heads, and those further from town were most likely to implement adaptations related to migration, asset depletion, change cropping patterns, or seeking out help, relative to reduced food consumption. Headship by women was associated more often with adaptation via asset depletion, changes in cropping patterns, or seeking out help, but not migration. The authors offered limited explanation of gender-related trends, aside from mentioning that female-headed households are generally viewed to be more vulnerable. There was no discussion of gender norms.

Conclusions

The authors concluded that farmers practiced a range of adaptations, and that farmers' ability to perceive climate risks and adapt was influenced by individual, household, and community capacity. They thus advocated for the development of improved infrastructure related to transportation, health, and education, as well as improved employment opportunities.

Reviewer comment

Relevant sections: possibly domestic responsibilities re: food security.

45. Scott, S., Arrieta, A., Kumar, N., Menon, P., & Quisumbing, A. (2020). Multidimensional predictors of common mental disorders among Indian mothers of 6- To 24-month-old children living in disadvantaged rural villages with women's self-help groups: A cross-sectional analysis. *PLoS ONE*, 15(6), 1-15. <https://doi.org/10.1371/journal.pone.0233418>

Research topic

Scott et al. (2020) examined the determinants of common mental disorders (CMDs) among mothers of young children in rural India.

Methodology

Using Self-Reporting Questionnaire data from over 1600 mother-baby pairs in rural areas across five Indian states, the authors examined four dimensions of women's lives: work (work type, time spent in labor, domestic and caretaking activities), agency (self-help group membership, decision-making, gender attitudes), health/nutrition (underweight, fertility, diet diversity, child illness), and household/environment (dependency ratio, wealth, food security, shocks, water, sanitation). They compared data from mothers with CMDs and those without, then used regression models to assess predictors.

Relevant findings

Scott et al. (2020) found that women were less likely to report CMD symptoms if they had heavier workloads, engaged in agricultural work rather than domestic work, were more involved in decision-making, and were members of self-help groups (especially those with microcredit activities). Various stressors were associated with an increased likelihood of CMD symptoms, including having a recently sick child, living in a food-insecure household, lacking improved sanitation, or experiencing recent livelihood-related shocks.

Conclusions

The findings indicated that rural mothers' resilience (in terms of mental health) was linked to their engagement in decision-making, their social capital and access to finance via self-help groups, and household poverty indicators.

Reviewer comment

Relevant sections: decision-making, social capital (SHGs), poverty. Note that this paper engages only superficially with 'food environment' issues but still seems relevant.

46. Serra, R., & Davidson, K. A. (2021). Selling Together: The Benefits of Cooperatives to Women Honey Producers in Ethiopia. *Journal of Agricultural Economics*, 72(1), 202-223. <https://doi.org/10.1111/1477-9552.12399>

Research topic

Serra et al. (2021) examined the impacts of collective action on Ethiopian women honey farmers' access to markets and economic outcomes.

Methodology

For this study, the authors drew on a survey of 916 women (332 cooperative members and 584 independent honey sellers) from two districts in Amhara and supplementary qualitative interview and focus group data from a separate study of similar individuals. They used relatively robust econometric methods—Coarsened Exact Matching (CEM) and regression analysis—to assess cooperative membership effects and factors contributing to women deriving greater or lesser benefits.

Relevant findings

In Ethiopia, Serra et al. (2021) reported, formal cooperative membership is generally limited to a single individual per household, leading to women's frequent exclusion—women are more likely to participate in informal self-help groups or farmer groups. The authors found that women's membership in a cooperative was associated with increased market prices, production levels, and, for wealthier members only, the share of honey product that was marketed. Possession of additional social capital, through self-help groups or farmer association membership, amplified these benefits for cooperative members. In interviews and group discussions, women reported increased mobility, control over household finances, heightened social status, and improved confidence.

Conclusions

The authors concluded that the benefits of cooperative membership for women were both economically and statistically significant, but that some aspects of women's identities (their wealth, distance to markets, and social capital, specifically) interacted to mediate the level of benefits derived from membership. They advocated removal of barriers to women's engagement in cooperatives more broadly.

Reviewer comment

Relevant sections: social capital, behavioral norms, income generation options.

47. Sibiya, N., Sithole, M., Mudau, L., & Simatele, M. D. (2022). Empowering the Voiceless: Securing the Participation of Marginalised Groups in Climate Change Governance in South Africa. *Sustainability (Switzerland)*, 14(12). <https://doi.org/10.3390/su14127111>Research topic

Research topic

Sibiya et al. (2022) examined community perceptions of how climate change adaptation is governed in South Africa. They argue that there is "an urgent need to move away from the instrumentalist approaches of participation and towards governance systems and structures which are capable of harnessing the ingenuity and energy of the poor to foster meaningful and transformative" adaptation (p. 2).

Methodology

The authors conducted 126 online surveys with community, church, and NGO leaders in four municipalities in eastern South Africa, as well as 13 semi-structured interviews with representatives of federal environmental agencies and municipal governments. Both quantitative and qualitative data were analyzed.

Relevant findings

Sibiya et al. (2022) found that most individuals surveyed were dissatisfied with the effectiveness of community initiatives, policies, and strategies related to adaptation, and community participation was very low (7%). Women respondents were less aware than men of community initiatives, policies, and strategies related to adaptation, despite women's heightened vulnerability linked to their lack of productive assets. The authors did not offer any explanation for this disparity in awareness.

Conclusions

Sibiya et al. (2022) concluded that adaptation governance structures were not inclusive and were inadequate in meeting the needs of the poor. Women's exclusion from decision-making around climate adaptation likely worsened the effectiveness and transformative potential of policies and strategies. The authors advocated for improved climate change education and agricultural skills development, alongside gender-sensitive climate adaptation governance that is more open to the poor and leverages indigenous knowledge.

Reviewer comment

Relevant sections: decision-making, access to information.

48. Singh, C., Solomon, D., & Rao, N. (2021). How does climate change adaptation policy in India consider gender? An analysis of 28 state action plans. *Climate Policy*, 21(7), 958-975.

<https://doi.org/10.1080/14693062.2021.1953434>

Research topic

Singh et al. (2021) assessed how gender mainstreaming was operationalized in sub-national climate change adaptation policies in India.

Methodology

The authors conducted a policy review of in 28 Indian states' climate change adaptation policies. They focused, first, on how gender and intersectional identities were recognized and conceptualized in each policy, and second, on how gender concerns were mainstreamed in approaches to build adaptive capacity and plan for adaptation.

Relevant findings

The authors first noted the framing of gender in India's National Action Plan on Climate Change, which references women's differential vulnerability: "With climate change, there would be increasing scarcity of water, reduction in yields of forest biomass, and increased risks to human health with children, women and the elderly in a household becoming the most vulnerable. With the possibility of a decline in availability of food grains, the threat of malnutrition may also increase. All these would add to deprivations that women already encounter and so in each of the adaptation programs, special attention should be paid to the aspects of gender" (cited p. 960). The authors found uneven engagement with gender across state plans; 43% of policies reviewed did not recognize gender, and five plans were deemed gender blind. Most of the remainder painted women as uniquely vulnerable victims of climate impacts, alongside other marginalized groups. These largely took a "gender-specific" approach, "targeting women and certain livelihoods, castes, and income groups through sectoral or livelihoods-based interventions" (p. 968) rather than addressing underlying structural inequalities. Only five policies depicted women as actors with agency to adapt and sought to address underlying inequalities through, e.g., more inclusive participation in public processes and integration of gender concerns into livelihoods programs. Half of the policies mentioned intersectional considerations in the form of differentiated livelihood opportunities, gendered divisions of labor (e.g., water and firewood collection), natural resource (e.g., land and water) access and use, and existing development deficits (e.g., marginalization and unequal access to education due to gender, caste).

Conclusions

The authors concluded that concepts around gender and adaptation were not thoroughly or consistently integrated in Indian state policies. While women were often treated as vulnerable victims, few policies acknowledged their agency and adaptive capacity or engaged with intersectionality beyond divisions of labor, resource access, and development deficits.

Reviewer comment

Relevant sections: income generation options, labor divisions, social capital. Note this is a policy review and not based on empirical data.

49. Tapia, M. A., Pulhin, J. M., Nelson, G. L. M., Predo, C. D., Peria, A. S., Peras, R. J. J., Lasco, R. D., Pulhin, F. B., & Evangelista, R. J. P. (2018). Involvement of women in farm decision-making and adaptive capacity to extreme events of farming households in Ligao City, Albay, Philippines. *Journal of Environmental Science and Management*, 21(2), 70-81.

https://doi.org/10.47125/jesam/2018_2/09

Research topic

Tapia et al. (2018), working in the Philippines, examined household decision-making structures related to farming and assessed whether there was a relationship between women's involvement in decision-making and households' adaptive capacity.

Methodology

The authors conducted roughly 300 household surveys in three landscape types, capturing decision-making around choice of food crops, cash crops, and tree species to be planted and farm production, as well as variables related to household adaptive capacity. The authors then used the Sustainable Livelihoods Framework to calculate households' adaptive capacity on the basis of the five capitals and used regression to evaluate which individual and household variables were associated with adaptive capacity.

Relevant findings

Tapia et al. (2018) found that husbands made sole decisions around farming in just over half of households, while joint decisions were made in roughly 1/3, depending on the decision in question. Women made sole decisions in only 13-16% of households, despite the fact that men worked off-farm in 30% of households. Women had the most say (either sole or joint) in decisions related to choice of cash crops, perhaps linked to their typical role in marketing produce. Interestingly, men's sole control over cash crop choice was associated with lower household adaptive capacity. The authors attribute these patterns to social norms that dictate men's control over decision-making, especially around trees (which they report require physical strength to manage), and possibly to women's substantially lower formal levels. They found that 96% of households had low adaptive capacity, which likely limited their ability to assess the potential causes of difference in capacity. They found no particularly clear patterns in predictors of adaptive capacity.

Conclusions

The authors concluded that decision-making around farming in this system is highly gendered and largely male-dominated. Even in areas where women had greatest responsibility, their decision-making authority was somewhat limited. They advocated for greater capacity for women.

Reviewer comment

Relevant sections: decision-making, differences in education; touches on division of labor and male migration.

50. Thinda, K. T., Ogundeji, A. A., Belle, J. A., & Ojo, T. O. (2020). Understanding the adoption of climate change adaptation strategies among smallholder farmers: Evidence from land reform beneficiaries in South Africa. *Land Use Policy*, 99(June), 104858. <https://doi.org/10.1016/j.landusepol.2020.104858>

Research topic

Thinda et al. (2020) examined smallholder farmers' adoption of climate adaptation strategies and their intensity in South Africa.

Methodology

The authors collected survey data from 183 farmers (39% female) in nine climate risk-prone municipalities and used a regression model to assess predictors of adaptation (as a dichotomous variable). Their data on adoption of adaptive strategies apparently came from open-ended questions, so any activities that a farmer described as an adaptation were included.

Relevant findings

Thinda et al. (2020) found that respondents practiced a range of adaptive strategies, including soil and water management, migration, use of improved and early maturing varieties, tree planting, and livestock reduction. They found that households with younger heads, female heads, non-farm income, and exposure to agricultural trainings, extension, and mobile phone based information sources were more likely to adapt. They offered no clear explanation of the gender-related pattern in uptake, aside from mentioning a separate study that showed female plot managers to be more likely to adopt specific sustainable intensification practices. Intensity of adaptation (seemingly how many adaptations a household practiced) was correlated with the age and education of the household head, farm size, access to credit, generation of non-farm income, and access to information. The authors did not further discuss gender.

Conclusions

The authors advocated for attention to education, farmer capacity building, and climate information.

Reviewer comment

Relevant sections: Potentially access to information, education, credit. No direct discussion of gender norms.

51. Thinda, K. T., Ogundeji, A. A., Belle, J. A., & Ojo, T. O. (2021). Determinants of Relevant Constraints Inhibiting Farmers' Adoption of Climate Change Adaptation Strategies in South Africa. *Journal of Asian and African Studies*, 56(3), 610-627. <https://doi.org/10.1177/0021909620934836>

Research topic

Thinda et al. (2021) assessed the factors associated with constraints on farmers' adaptation strategies.

Methodology

The authors implemented a survey with 183 farmers (39% women) in nine district municipalities across four provinces of South Africa, then used descriptive statistics and a multivariate probit model to identify the determinants of farmers' reported constraints on adaptation.

Relevant findings

Thinda et al. (2021) found that farmers' three most commonly reported constraints to adaptation were a lack of information about climate change (reported by 90%), a lack of knowledge about climate change (reported by 77% of farmers), and a lack of capital (reported by 74%) (all treated as dichotomous variables). Older farmers, members of farmer organizations, households with non-farm income, and male-headed households were found less likely to face knowledge-related constraints. Women's limited access to land and other productive resources was offered as an explanation for this gender disparity, although the link between resource ownership and women's knowledge is unclear. They found that information-related constraints to adaptation were associated with farmers lacking training, access to extension, non-farm income, and access to radio. Constraints related to capital were experienced most by farmers who lacked access to extension, training, and radio communications. The authors do not engage directly with gender norms, but their findings around extension and training may be relevant.

Conclusions

The authors advocated for expanded access to climate information and credit.

Reviewer comment

Relevant sections: possibly access to information, extension. No direct engagement with gender norms.

52. Thoai, T. Q., Rañola, R. F., Camacho, L. D., & Simelton, E. (2018). Determinants of farmers' adaptation to climate change in agricultural production in the central region of Vietnam. *Land Use Policy*, 70(October 2017), 224-231. <https://doi.org/10.1016/j.landusepol.2017.10.023>

Research topic

Thoai et al. (2018) examined the impact of assets on farmers' adaptation decisions in north-central Vietnam.

Methodology

The authors used a survey of 400 individual farmers through which they collected data on individual's access to the five capitals and their self-reported adaptive behaviors. They then used a logit model to identify factors associated with adaptation in general (as a dichotomous variable) and a multivariable model to explore factors associated with different adaptive behaviors.

Relevant findings

Thoai et al. (2018) found that the household's engagement in a climate change training and larger farm size were predictors of adaptation, but gender was only significant in relation to the decision to change crop varieties (which women were more likely to do than men). Access to credit was significantly associated with adjusting farming calendars and monitoring weather forecasts, and a few other predictors emerged as significant in the case of specific adaptive behaviors, but there were not many clear signals

about the drivers of adaptive behaviors. The authors did not address the role of gender norms or offer explanations for the patterns they found.

Conclusions

The authors concluded that farmers in this context were heavily affected by climate change and were adapting in various ways, primarily by monitoring weather forecasts and changing crop varieties. They advocated for expanded education and training courses tailored to poorer and less experienced farmers and better support for smallholders in the region.

Reviewer comment

No discussion of gender norms.

53. Thomas, A. S., Mangubhai, S., Vandervord, C., Fox, M., & Nand, Y. (2019). Impact of Tropical Cyclone Winston on women mud crab fishers in Fiji. *Climate and Development*, 11(8), 699-709. <https://doi.org/10.1080/17565529.2018.1547677>

Research topic

Thomas et al. (2019) assessed the impacts of a tropical cyclone on mud crab fishers (who are largely women) in Fiji to understand their vulnerability, livelihood impacts of the cyclone, and barriers to adaptation.

Methodology

The authors used a pre-cyclone value chain assessment survey (n = 109) and a post-cyclone socioeconomic survey with identical and complementary questions, contacting 55 of the previous fishers and adding a few 'new' fishers for a total of 68 individuals. In addition to comparing pre- and post-cyclone responses, they used regression models to identify predictors (e.g., demographics, livelihoods information, perceptions of cyclone impacts) of continuing to harvest mud crabs post-cyclone.

Relevant findings

Thomas et al. (2019) found that more than half of their respondents had stopped harvesting mud crabs two to three months after the cyclone, and some faced physical obstacles to accessing both mangroves and crab markets. Social norms did emerge as likely contributors to women's vulnerability; in fact, although most fishers reported equipment losses, women generally experienced less damage because their equipment was simpler than men's (i.e., hand-harvesting rather than boats). The authors suggested that women did, however, likely face greater difficulties engaging in alternative livelihoods post-cyclone due to their limited education, as well as challenges in accessing necessary new equipment due to relatively limited access to credit. Likely due to the small sample size, the authors were unable to achieve a good fit with regression models, so could not identify predictors of continued crab fishing, and could not quantitatively compare behaviors across genders.

Conclusions

The authors concluded that vulnerability in this context was influenced by both biophysical and socioeconomic factors. They advocated for gender-sensitive government support, better regulation of fisheries, protection of vulnerable areas, and improved monitoring.

Reviewer comment

Relevant sections: touches on education, access to finance, access to diverse livelihoods.

54. Vogliano, C., Raneri, J. E., Coad, J., Tutua, S., Wham, C., Lachat, C., & Burlingame, B. (2021). Dietary agrobiodiversity for improved nutrition and health outcomes within a transitioning indigenous Solomon Island food system. *Food Security*, 13(4), 819-847. <https://doi.org/10.1007/s12571-021-01167-7>

Research topic

Vogliano et al. (2021) explored the relationship between dietary agrobiodiversity and nutrition outcomes in the Solomon Islands.

Methodology

The authors conducted nutrition surveys with thirty women responsible for food preparation (often women of reproductive age who are generally vulnerable to malnutrition) in households in a single village, also capturing nutrition-related knowledge, attitudes, and practices, including barriers to accessing healthy food. They also organized fourteen focus group discussions to understand the historical trajectory of local climate, food systems, and diets.

Relevant findings

The authors found several indicators of poor dietary quality and some evidence that increased dietary diversity was associated with lower body fat percentages. It was also clear that indigenous knowledge and food systems were undermined by changing food demands and increasing reliance on imports and processed foods. Processed foods were preferred not only for taste and low cost, but for ease and speed of cooking—relevant to women responsible for both fuel collection and food preparation. The authors do not discuss gender beyond this.

Conclusions

The authors concluded that there was evidence that modern food systems oriented around heavily processed fuels were fueling poor health outcomes. They advocated for a coordinated nutrition strategy to build a more sustainable food system.

Reviewer comment

Relevant sections: potentially domestic responsibilities. This paper is a bit tangential.

55. Witinok-Huber, R., & Radil, S. M. (2021). Introducing the Local Agricultural Potential Index: An approach to understand local agricultural extension impact for farmer adaptive capacity and gender equity. *World Development Perspectives*, 23, 100345. <https://doi.org/10.1016/j.wdp.2021.100345>

Research topic

Witinok-Huber & Radil (2021) explored the role of extension services in farmers' adaptation processes.

Methodology

The authors adopted a conceptual framework establishing adaptive capacity as a function of access and agency. Using data from a survey of 352 farmers and focus group discussions in north-central Liberia, they built the multidimensional Local Agricultural Potential Index, using WEIA as a foundation. Their index was based on four domains, two touching on access and two on agency: 1) resource access (including extension, loans/credit, and land; 2) membership and leadership in farmer-based and community groups; 3) household power (including decision-making and money management); and 4) time allocation toward productive and domestic activities. They then used ANOVA models to assess the influence of gender and geography on adaptive capacity scores.

Relevant findings

The authors found that on average, women had significantly lower scores across all access and agency domains than men, and significantly lower overall index scores. This "reflects [that women have] less access to ICT, leadership opportunities, and credit/loans, as well as agency or the social power to make decisions at home and in the community when compared to their male counterparts" (p. 7). They noted that unequal ICT ownership is often tied to gender norms in Liberia and that "deep-rooted cultural and religious beliefs of women's value and roles in the home and community further limit their opportunities to make critical decisions, own land, be educated, and involved in commercial agriculture" (p. 4).

Conclusions

The authors concluded that gender as well as location (with accompanying variation in social norms and biophysical conditions) affect individuals' needs and constraints, making it important to tailor extension services to localized, intersecting needs.

Reviewer comment

Relevant sections: resource access (information/extension, land), social capital, labor burdens, decision-making.

56. Wood, A. L., Rivers, L., Sidbé, A., & Ligmann-Zielinska, A. (2021). Decision-making capacity to address climate-induced food insecurity within women-led groups in Southern Mali. *Climatic Change*, 164(3-4). <https://doi.org/10.1007/s10584-021-03003-5>

Research topic

Wood et al. (2021) evaluated perceptions of climate change and the inter- and intra-gender power dynamics related to adaptation in polygynous Malian farming households.

Methodology

The authors interviewed 41 farmers (20 women, 21 men) in six communities to discuss household food systems.

Relevant findings

Wood et al. (2021) argued that in polygynous households, senior women often have decision-making authority over younger women, meaning these groups of women “are navigating different climate-linked realities, despite occupying the same spaces” (p. 29). Through their study they found that while elder men engaged most in decision-making and young women least, young women took on the most ‘actions’ – i.e., tasks to support household food system operations. Under climate stress, women’s (especially young women’s) task loads were amplified to mitigate food insecurity, although women had limited authority to alter farming strategies. Elder women exerted decision-making control over household tasks and resources that elder men allotted, and that young women in turn executed. Young women appeared to have a greater diversity of off-farm livelihoods available to them than elder women. However, unmarried women could not control the money they earned (it went to their mothers), which limited their ability to contribute directly to household adaptation. Elder women, although subject to the broader authority of elder men, sometimes engaged in agricultural collectives or contributed personal earnings to the household food system in times of shortage. Collectives increased their perceived self-efficacy and resilience, allowing them control over agricultural decisions that they lacked at home.

Conclusions

The authors concluded that power dynamics within households meant that women faced differing constraints depending on their age and marital status. Young women were most vulnerable due to their limited decision-making authority.

Reviewer comment

Relevant sections: decision-making, social capital, labor burdens (domestic work), control over finances, income generation options, intersectionality.

57. Wouterse, F., Andrijevic, M., & Schaeffer, M. (2022). The microeconomics of adaptation: Evidence from smallholders in Ethiopia and Niger. *World Development*, 154, 105884. <https://doi.org/10.1016/j.worlddev.2022.105884>

Research topic

Wouterse et al. (2022) examined the drivers and welfare outcomes associated with adaptive strategies in Ethiopia and Niger using the IPCC framework for risk, exposure, and vulnerability.

Methodology

The authors drew on the latest available Living Standards Measurement Survey data from 2,136 rural households in Niger and 3,680 in Ethiopia. They then used cluster analysis to develop farm household typologies and regression models to identify predictors and outcomes associated with adaptations.

Relevant findings

Wouterse et al. (2022) found that access to resources (namely land and livestock) and formal education, as well as perceived climate risk, were associated with adaptation. Education in particular was tied to lower vulnerability, less food insecurity, more on-farm adaptation, and livelihood diversification. In both countries, female-headed households were less likely to practice on-farm adaptive strategies, and in Niger, female-headed households were less food secure. However, female-headed households were in some cases more diversified in terms of livelihoods, and thus less exposed to climate risk. Although the authors did not

explore relationships between indicators of adaptive capacity and gender of the household head, they acknowledged that access to key resources is often highly gendered. They also found that perceptions of risk and exposure to stressors varied according to sociodemographic variables such as gender. They attributed this to men and women's frequent occupation of different spheres, which led them to differentially weigh risks, e.g., men have been found to prioritize risks to farm activities while women prioritized risks to the wellbeing and education of household members. The authors did not find close associations between adaptation and food security outcomes, but did identify "spatial poverty traps" (i.e., especially vulnerable micro-regions).

Conclusions

The authors concluded that education and assets played key roles in shaping households' adaptive capacity. They advocated for gender-inclusive education interventions to increase both adaptive capacity and awareness of risk, cash transfers to bolster adaptive capacity of resource-poor households, and attention to areas prone to food insecurity.

Reviewer comment

Relevant sections: education, access to resources/land, behavioral norms / division of labor.

58. Yorick, R., Khudonazarov, F., Gall, A. J., Pedersen, K. F., & Wesson, J. (2021). Volunteer community health and agriculture workers help reduce childhood malnutrition in Tajikistan. *Global Health Science and Practice*, 16, S137-S150. <https://doi.org/10.9745/GHSP-D-20-00325>

Research topic

Yorick et al. (2021) examined the health impacts of nutrition and WASH trainings for community health workers and community agriculture workers in Tajikistan.

Methodology

Trained community workers (89% women) paid household visits in their areas, where malnutrition among women and children was a key concern. The authors conducted panel surveys of agricultural practices and household practices measuring, among other things, sanitation, hygiene, diet, and nutrition outcomes.

Relevant findings

Yorick et al. (2021) found that the activities of trained community workers led to statistically significant improvements in several indicators of women's and children's health—especially children's dietary quality and women's WASH practices. The researchers also learned that it was critical to engage older women within households first during this outreach, as older women made most household decisions (especially while men were away at work), while young women handled most domestic responsibilities. Without buy-in from older women, young women were not usually permitted to participate in trainings or necessarily adopt improved behaviors (e.g., eating enough and spending adequate time breastfeeding). Community volunteers were able to continue in part because they earned modest stipends, making this work acceptable within the household, but still struggled at times to enter "difficult" households (allegedly due to capacity for communication). The authors do not engage further with questions of social norms, including their role in women's health and nutrition outcomes.

Conclusions

The authors concluded that community health workers and community agricultural workers can have significant impacts on health and nutrition outcomes, but that these programs need careful design and adequate support to maintain.

Reviewer comment

Relevant sections: intersectionality, access to information/training, decision-making.

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Gender Equality

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